Performance Analysis of Digital Modulation and Detection Techniques

Date Performed:
Expt. No.: 1
Reg. No.:
Name:

1 Objective

To conduct an experiment and study the performance of digital modulation and demodulation technique with the aid of NI-USRP 2920/2901.

2 Equipment / Tool(s) Required:

- NI -USRP 2920/2901
- VERT 900: 824-960 MHz
- 1710-1990 MHz dual-band cellular and ISM vertical antenna

3 Theory

3.1 Modulation:

Digital modulation consists of mapping a sequence of bits to a (finite) set of analogue waveforms which are suitable for transmission over a particular channel in which symbols are modulated onto pulse-shapes. The source for digital modulation is a sequence of bits b[n]. The bit sequence is passed to the physical layer of a radio by higher layers. The sequence of bits is processed by the symbol mapping block to produce a sequence of symbols s[n], each value is a complex number that comes from a finite set of symbols called the constellation.

$$c = \{s_0, \ldots, s_{M-2}, s_{M-1}\}$$

The size of the constellation, or cardinality, is denoted $|C| = M$ where M is the number of symbols in the constellation. For practical implementations $M = 2^b$ where b is the bits per symbol. The symbols are pulse shaped and scaled by a factor of$\sqrt{Ex}$ to produce complex base band signals.

$$\sqrt{E_x} \sum_{\infty}^{n=-\infty} \delta[n]g$$

The scaling factor E_x is used to model the effect of adding energy or power to x(t). The scaling factor will be added typically in the RF by controlling the gain of the transmit power amplifier. The pulse-shape is given by the function $g_{tx}(t)$. To preserve energy, the pulse-shaping function is assumed to be normalized such that $\int_{-\infty}^{\infty} |g(t)|^2 dt$. The symbol period is given by T. The symbol rate is 1/T. We refer to it as a complex pulse amplitude signal because complex x(t) symbols modulate a succession of pulses given by the function $g_{tx}(t)$. Typically, in wireless systems this is measured in kilo-symbols per second or mega-symbols per second. Be careful to note that 1/T is not necessarily the bandwidth of x(t), which depends on $g_{tx}(t)$. The signal is complex because the resulting wireless signal is up converted to x(t) on a carrier frequency. After up conversion, the resulting pass band. After up conversion, the resulting pass band.

$$x_p(t) = R_e\{x(t)\}cos(2\pi f_c t) - Imx(t)sin(2\pi f_c t)$$

Up conversion is performed in the analogue hardware. The signal will be $x_p(t)$ launched from the transmit antennas into the propagation environment.

3.2 Demodulation

When a signal propagates through the wireless channel, it will be affected by noise and other types of channel distortion. The type of distortion impact the architecture of the receiver and the design of the signal processing algorithms. The AWGN communication channel is a good model for the impairments due to thermal noise present in any wireless communication system.

$$z(t) = x(t) + c(t)$$

where x(t) is the complex baseband signal, v(t) is the AWGN, and z(t) is the observation. The assumption that $v(t)$ is AWGN means the noise is independent and identically distributed (i.i.d.) complex Gaussian random variable. The effective noise temperature is a function of the ambient temperature, the type of antennas, as well as the material properties of the analogfront end. Under the assumed transmitted structure, the optimum receiver structure involves matched filtering, sampling at the symbol rate, and detection The receive- d signal z(t), which is assumed to be bandlimited, is sampled with a sampling rate of $1/T_z$. The sampling rate should be such that Nyquist is satisfied. Assume that $T_z = T/M$ where M is the integer oversampling factor. The received signal is filtered by $g_{rx}[n] = g_{rx}(nT_z)$ where $g_{rx}(t) = g_{tx}(-t)$ is the matched filter.

4 Flowchart

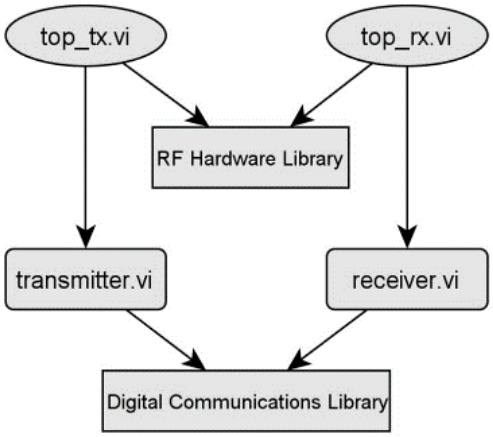


Figure 1: Hierarchy of code used in lab[1]

5 Block Diagram

5.1 Transmitter.vi

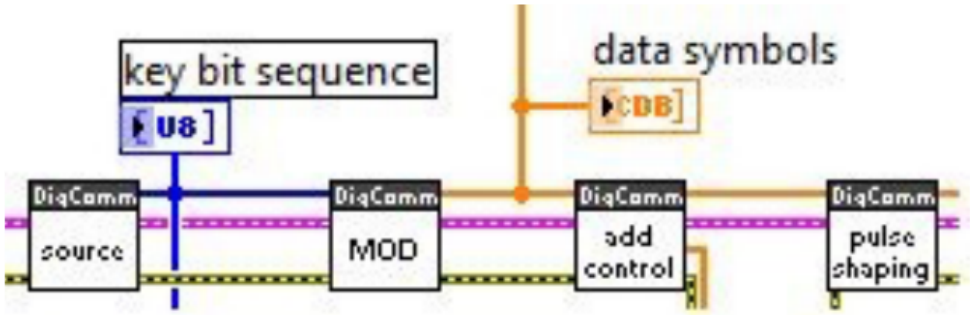


Figure 2: Block diagram of transmitter.vi[1]

5.2 Receiver.vi

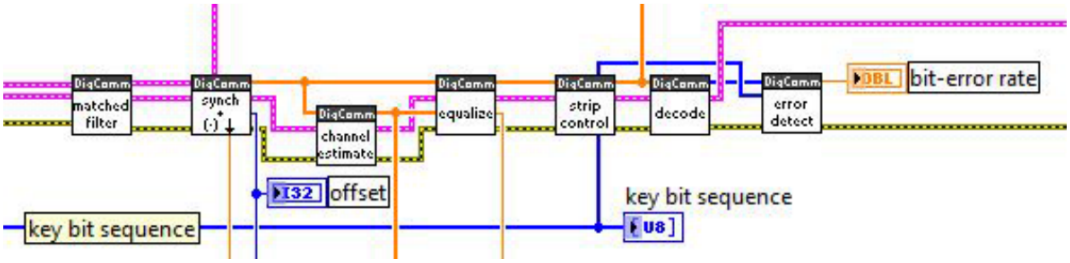


Figure 3: Block diagram of receiver.vi[1]

6 Procedure

Construct transmitter.vi and receiver.vi files with following input parameters.

Modulate.vi-maps it sequences to symbols using appropriate constellation; modifies symbol energy of output symbol sequence			
INPUTS	Input bit sequence	Array of U8	Input bit sequence (each array element takes on value 0 or 1)
	Modulation type	string	Determines type of constellation map (i.e. BPSK or QPSK)
	Symbol energy	double	Determines the per-symbol energy of the output sequence
OUTPUTS	Output symbol	Array of CDB	Appropriately modulated output symbols.

Decode.vi-performs ML detection on a given symbol sequence			
INPUTS	Input symbols	Array of CDB	Input symbols to be decoded
OUTPUTS	Estimated bit sequence	Array of U8	Decoded output bit sequence

Construct top_tx.vi, top_rx.vi with the following parameters

- Packet length=100 bits
- Modulation type= M-ary QAM
- Start the top level of your transmitter (i.e, run top_tx.vi)
- Wait until the transmitting indicator is lit.
- Start the top level of your receiver (i.e, run top_rx.vi).

- Open the front panel of top_rx.vi to see the constellation of the received signal.
- The key bit sequence is transmitted in the form of a queue to the receiver for error detection and control for channel model parameters in transmitter.vi is created in top_tx.vi.
- These parameters are used to apply a desired channel model to the transmitted signal.

7 Results and Conclusions

7.1 4-QAM

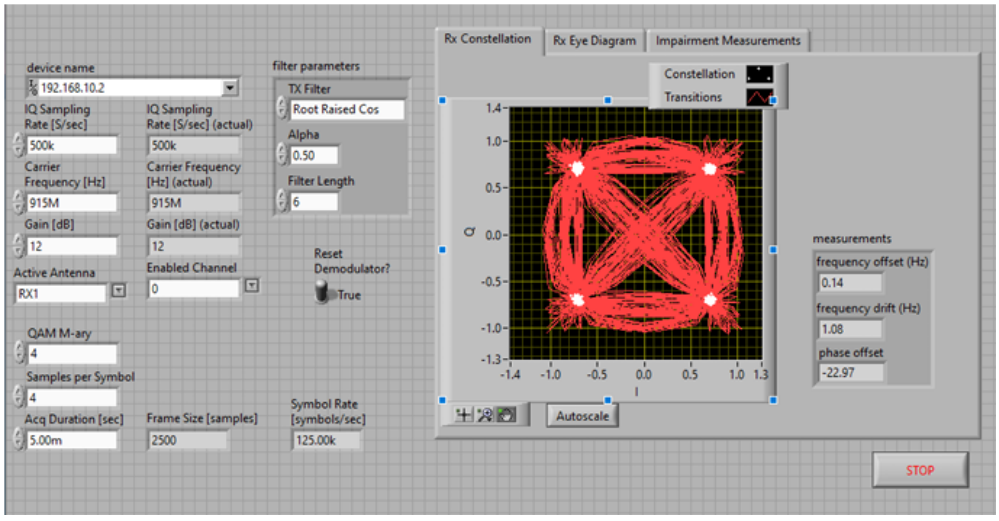


Figure 4: Constellation diagram

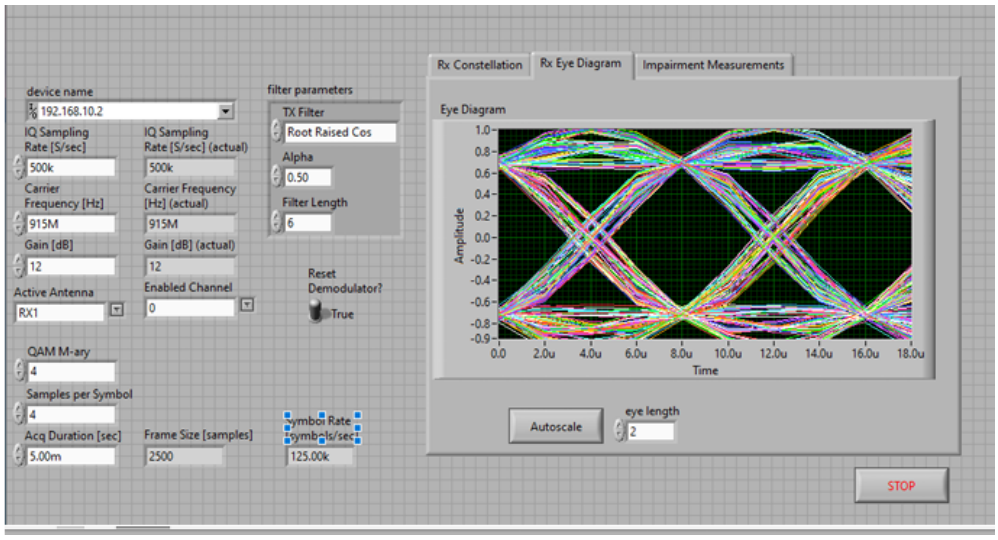


Figure 5: Eye diagram

7.2 16-QAM

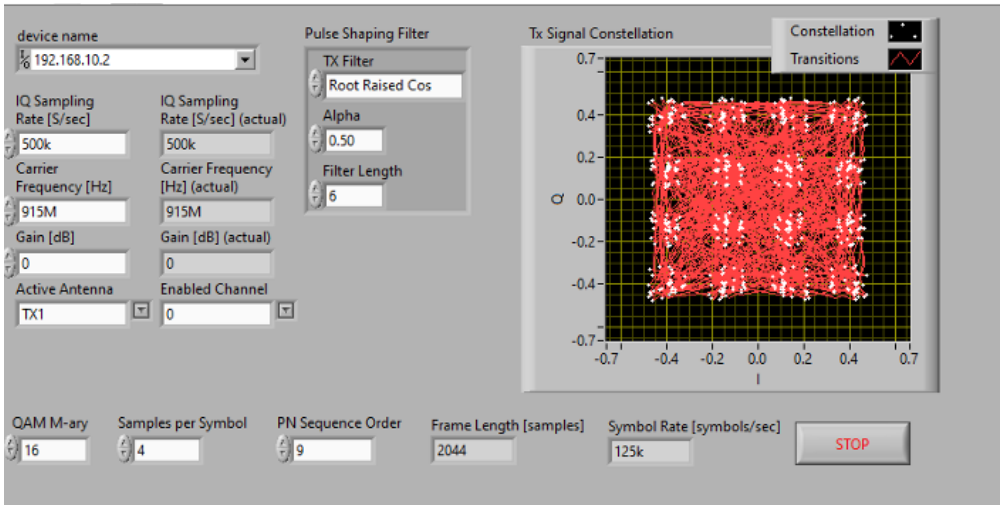


Figure 6: Constellation diagram

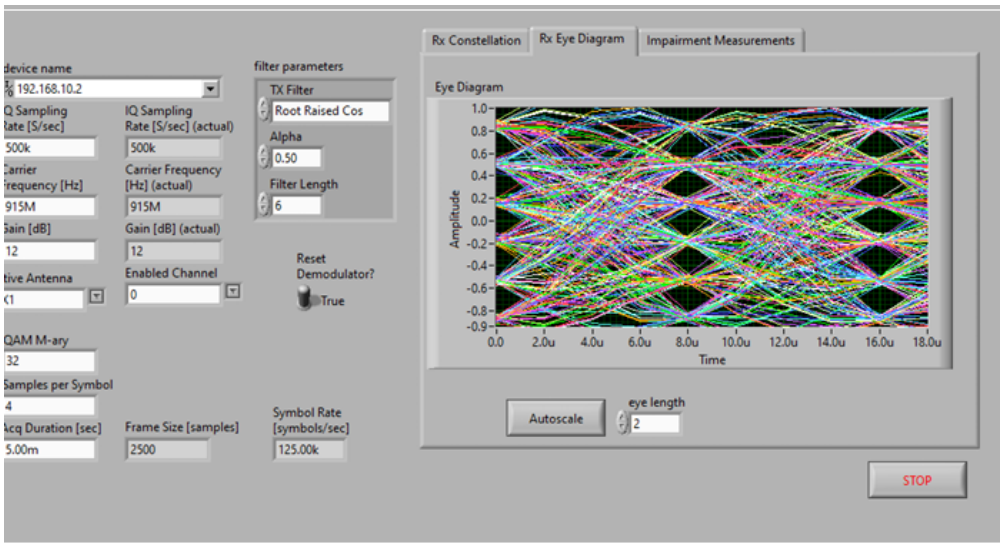


Figure 7: Eye diagram

7.3 32-QAM

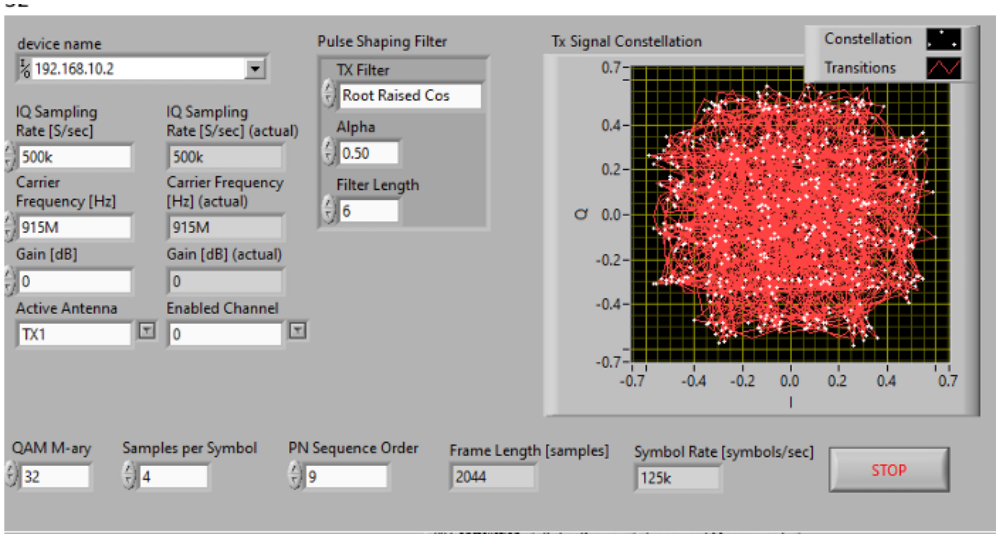


Figure 8: Constellation diagram

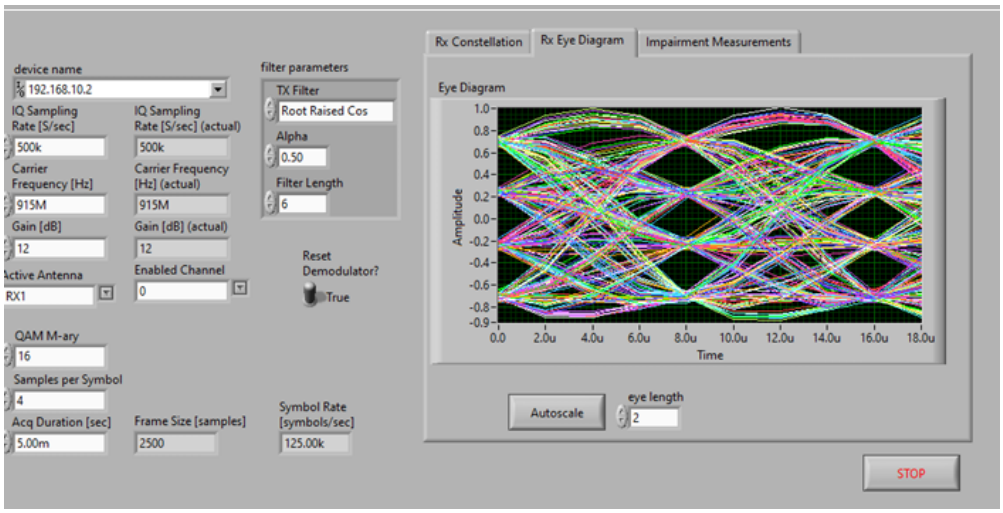


Figure 9: Eye diagram

8 Results and Discussion

References

[1] H. Robert W, “Digital communications-Physical layer exploration lab using the NI USRP platform,”

Pulse Shaping and Matched Filtering Analysis

Date Performed:
Expt. No.: 2
Reg. No.:
Name:

1 Objective

To conduct an experiment and study the pulse shaping and matched filtering with the aid of NI - USRP 2920/2901.

2 Equipments/Tool(s) Required

- NI - USRP 2920/2901
- VERT 900: 824-960 MHz, 1710-1990 MHz dual-band cellular and ISM vertical antenna
- LABVIEW

3 Theory

Pulse shaping is the process of changing the waveform of transmitted pulses. Its purpose is to make the transmitted signal better suited to its purpose or the communication channel, typically by limiting the effective bandwidth of the transmission. By filtering the transmitted pulses this way, the intersymbol interference caused by the channel can be kept in control. In RF communication, pulse shaping is essential for making the signal fit in its frequency band. Typically pulse shaping occurs after line coding and modulation.
The matched filter is the optimal linear filter for maximizing the signal-to-noise ratio (SNR) in the presence of additive stochastic noise. Matched filters are commonly used in radar, in which a known signal is sent out, and the reflected signal is examined for common elements of the out-going signal.
A digital signal cannot be transmitted over a communication channel. The digital signal needs to be mapped to an analog waveform that can easily be transmitted over a channel.

The transmitter creates a complex pulse amplitude modulated signal of the following form:

$$x(t) = \sqrt{E_x} \sum_{n=-\infty}^{\infty} s[n] g_{tx}(t - nT)$$

Conceptually, the mapping of bits to waveform can be viewed in two steps:

a. **Source bits to symbols** - This involves mapping the bits emitted by the (digital) source to (possibly complex) symbols on a constellation C determined by the modulation scheme.
b. **Symbols to pulses** - This involves creating an analog pulse train to be transmitted over the wireless channel, using complex symbols generated in the first step. This is done using a pulse-shaping filter $g_{tx}(t)$ at the transmit side.

The receiver processing depends on the type of distortion in the communication channel. Assuming an additive white Gaussian noise (AWGN) channel, the receiver performs analogous operations at the transmitter.

a. **Pulses to symbols** - In this step, the symbols are recovered from the received pulses using a matched filter.

b. **Symbols to bits** - This involves mapping the symbols received to bits (depending on the modulation scheme chosen), using some form of a detection technique.

4 Flowchart

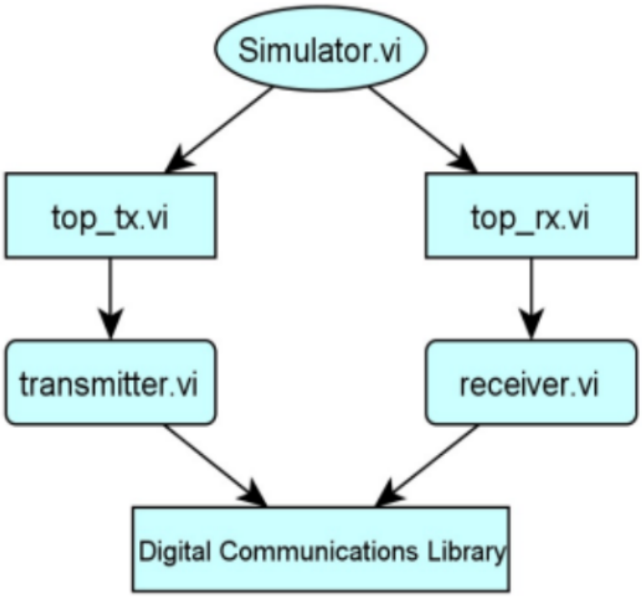


Figure 1: Simulator hierarchy [1]

5 Block Diagram

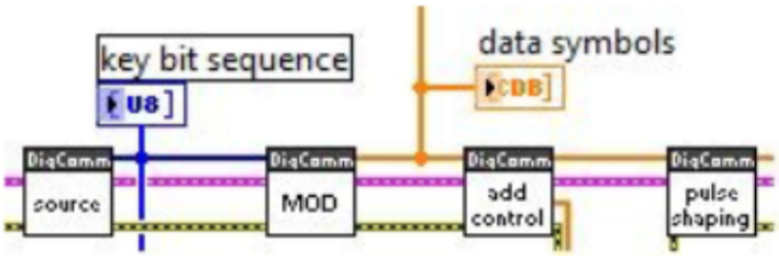


Figure 2: Transmitter.vi [1]

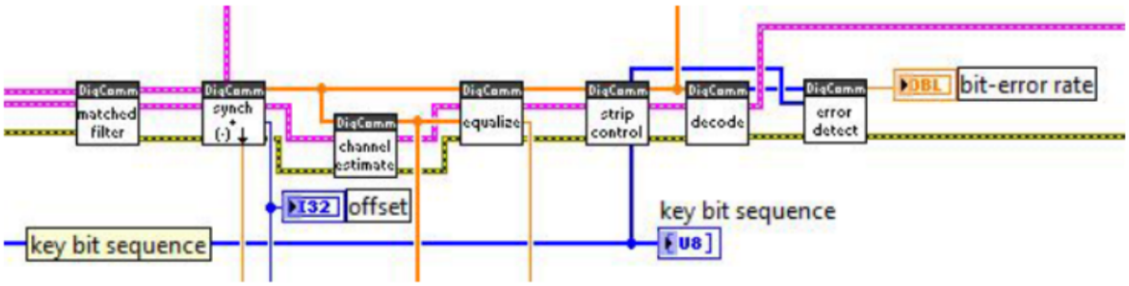


Figure 3: Receiver.vi [1]

6 Procedure

Description of pulse shaping.vi:

pulse_shaping.vi - oversample input and apply transmit pulse shaping filter			
INPUTS	Input bit Sequence	Array of U8	Input bit sequence (each array element takes on value 0 or 1)
OUTPUTS	Output Symbol	Array of CDB	Appropriately modulated output symbols.

Description of matched filtering.vi:

matched_filtering.vi - apply matched filter corresponding to pulse shaping parameters			
INPUTS	Input complex waveform	IQ waveform cluster	Received Sequence
OUTPUTS	Estimated bit sequence	Array of U8	Decoded output bit sequence

Construct top_tx.vi, top_rx.vi with the following parameters

- Packet length =200 bits
- Modulation type = QPSK
- Pulse shaping filter = Root raised cosine
- Filter parameter = 0.5
- Filter length = 8
- Capture time = 2msec

7 Results and Conclusions

a. α=0.2

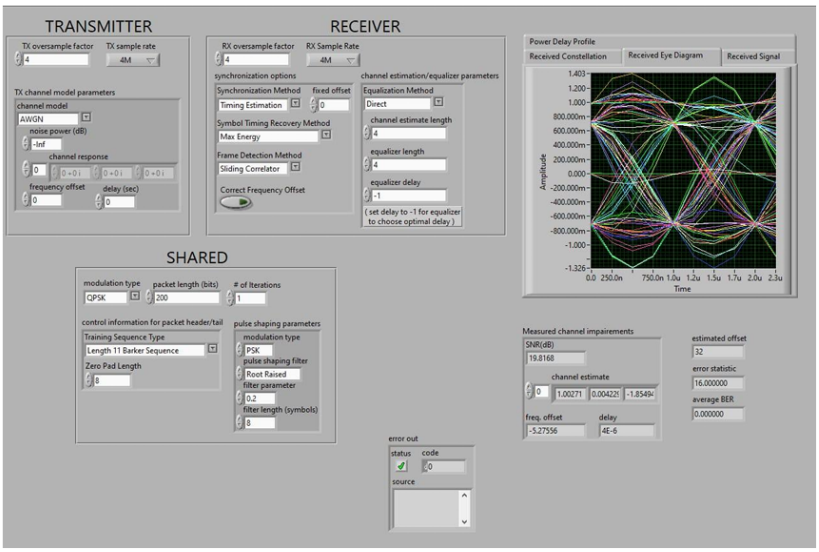


Figure 4: Eye diagram

b. α=0.4

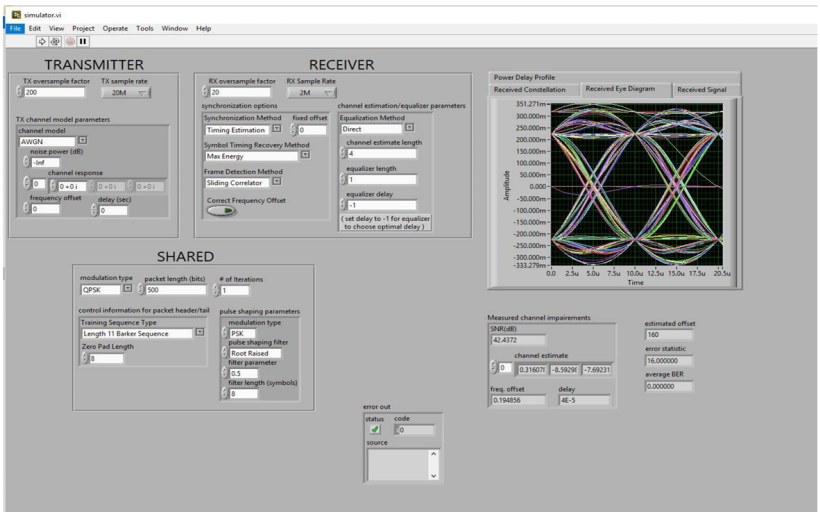


Figure 5: Eye diagram

c. α=0.6

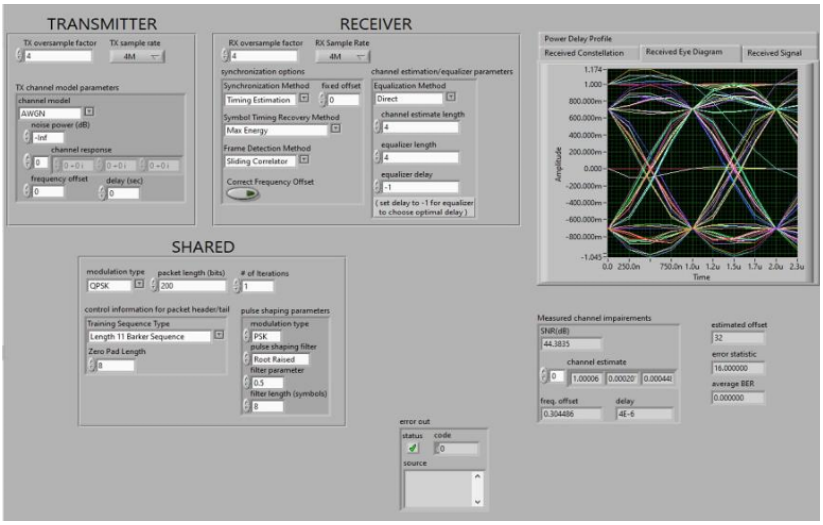


Figure 6: Eye diagram

d. α=0.8

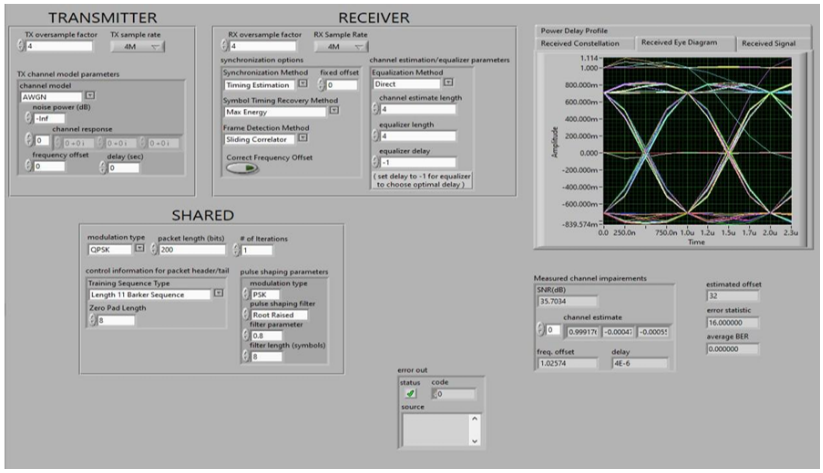


Figure 7: Eye diagram

8 Results and Discussion

References

[1] H. Robert W, “Digital communications-Physical layer exploration lab using the NI USRP platform.”

Demonstration of Synchronization

Date Performed:
Expt. No.: 3
Reg. No.:
Name:

1 Objective

To conduct an experiment and study the synchronization technique with the aid of NI - USRP 2920/2901.

2 Equipments/Tool(s) Required

- NI - USRP 2920/2901
- VERT 900: 824-960 MHz, 1710-1990 MHz dual-band cellular and ISM vertical antenna
- LABVIEW

3 Theory

The wireless communication channel is not well modelled by simple additive white Gaussian noise. A more realistic channel model also includes attenuation, phase shifts, and propagation delays. Perhaps the simplest channel model is known as the frequency flat channel. The frequency flat channel creates the received signal

$$z(t) = \alpha e^{j\phi} x(t - \tau_{d)} + v(t)$$

where α is an attenuation, ϕ is a phase shift, and τ_d is the delay.

The objective of this lab is to correct for the delay caused by τ_d in discrete time. The approach will be to determine an amount of delay $\hat{k}$ and then to delay the filtered received signal by $\hat{k}$ prior to down sampling. This will modify the receiver processing as illustrated in Figure 1. Two algorithms will be implemented for symbol synchronization in this lab: the maximum energy method and the Early Late gate algorithm. The maximum energy method attempts to find the sample point that maximizes the average received energy. The early–late gate algorithm implements a discrete-time version of a continuous-time optimization to maximize a certain cost function. For the best performance, both algorithms require that there is a lot of oversampling (i.e., M is large).

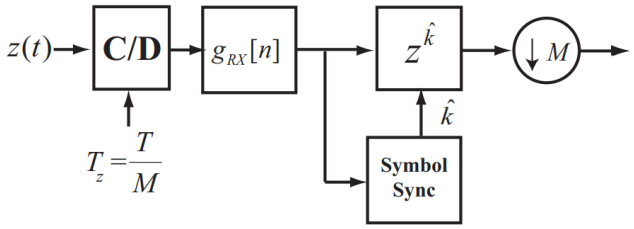


Figure 1: Receiver with symbol synchronization after the digital matched filtering. [1]

4 Flowchart

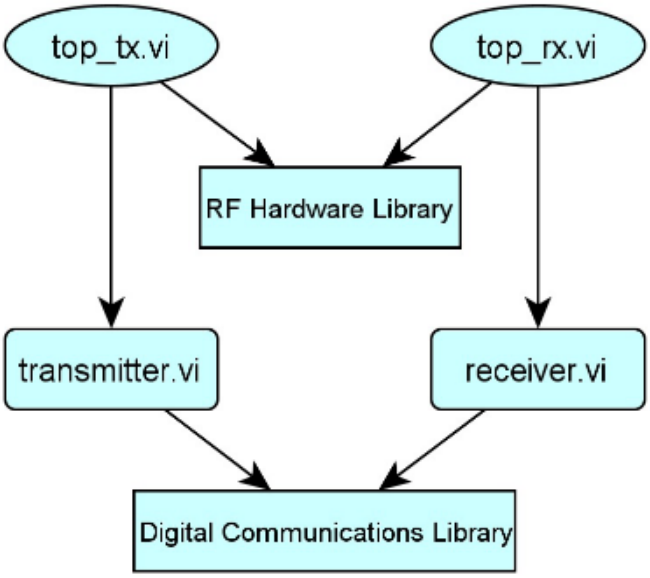


Figure 2: Hierarchy of code used in lab [1]

5 Block Diagram

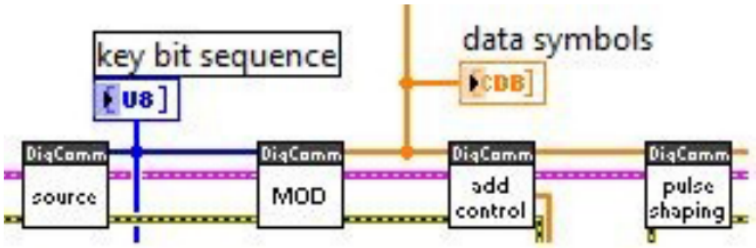


Figure 3: Transmitter.vi [1]

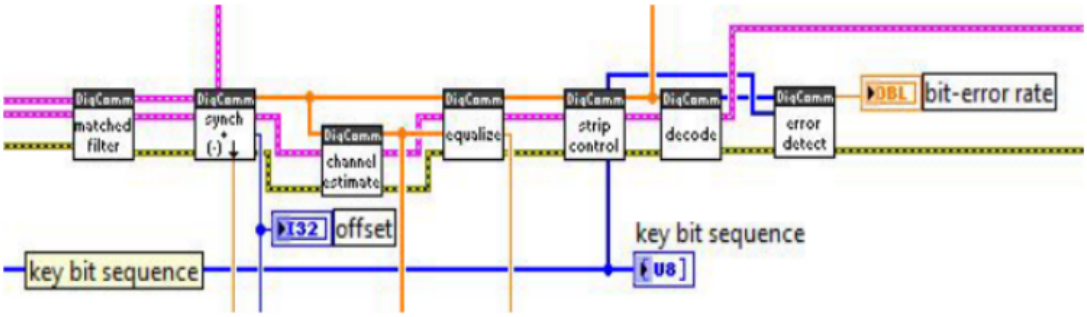


Figure 4: Receiver.vi [1]

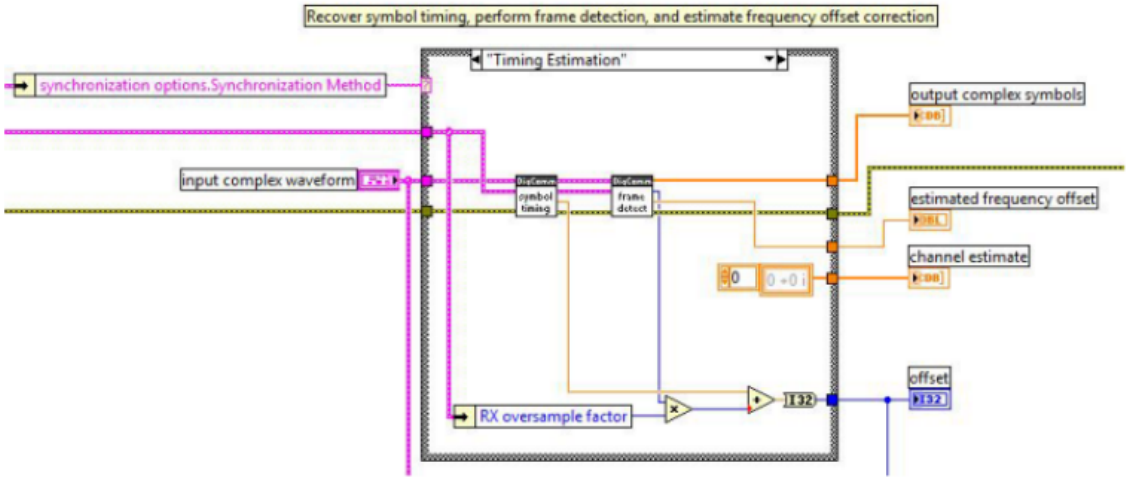


Figure 5: Synchronization.vi [1]

6 Procedure

Description of pulse shaping.vi:

align MaxEnergy.vi - compute sampling offset by maximizing $J_{approx}[k]$ and correct for offset			
INPUTS	input complex waveform	IQ Waveform cluster	received sequence after matched filtering
OUTPUTS	output complex waveform	IQ Waveform cluster	received sequence after symbol timing recovery
	offset	132 (integer)	computed offset

Description of matched filtering.vi:

align ELgate.vi - estimate timing offset by minimizing $J_{\delta}[k]$ and correct for offset			
INPUTS	input complex waveform	IQ Waveform cluster	received sequence after matched filtering
OUTPUTS	output complex waveform	IQ Waveform cluster	received sequence after symbol timing recovery
	Offset	132 (integer)	computed offset

7 Results and Conclusions

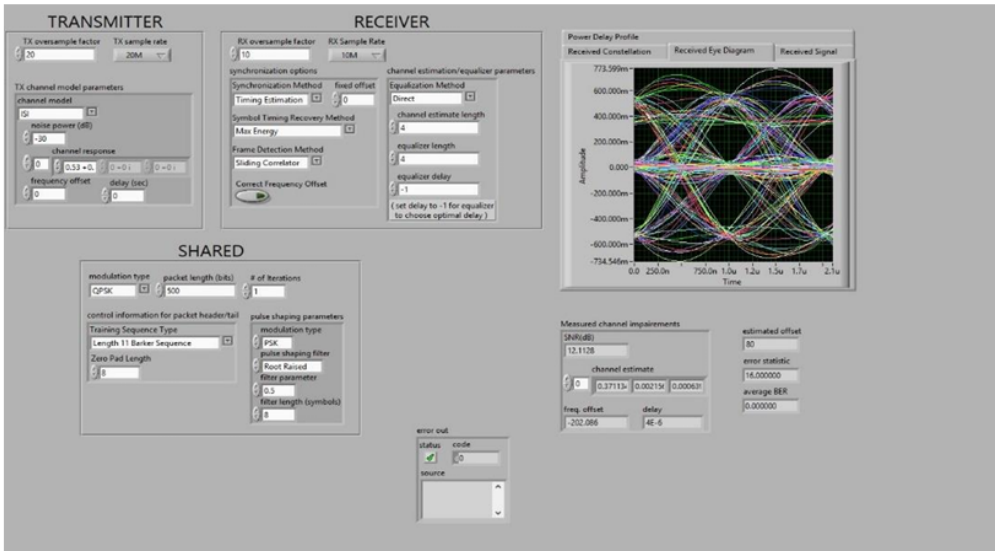


Figure 6: Received Eye Diagram using Symbol Timing Synchronization

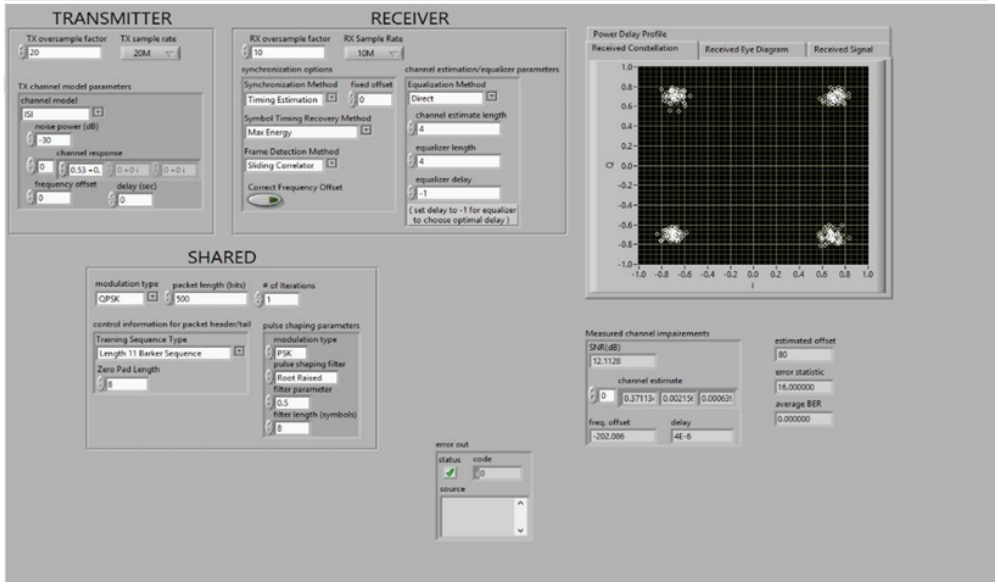


Figure 7: Received QPSK Constellation using Symbol Timing Synchronization

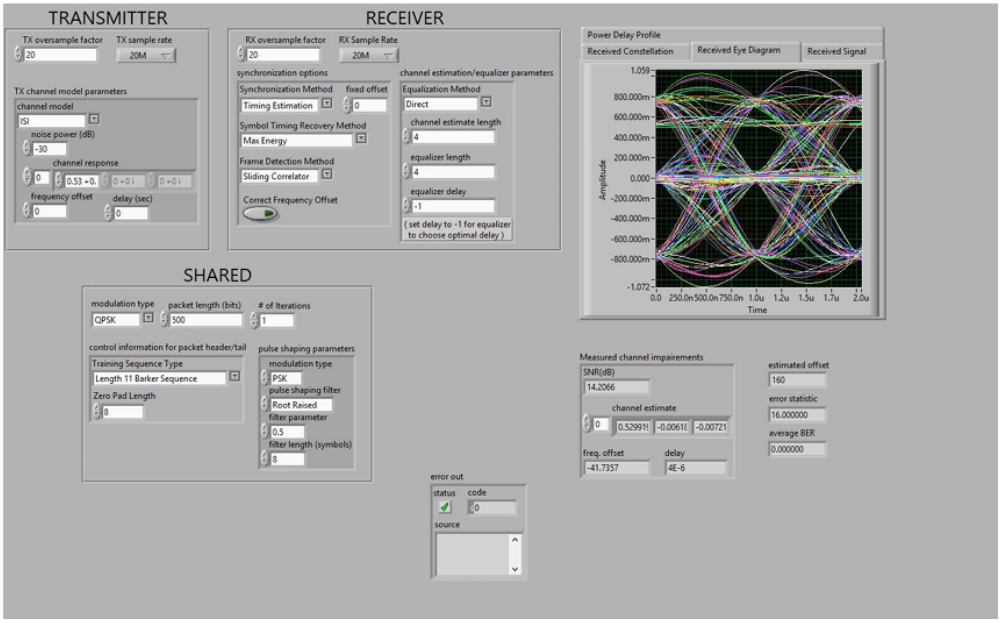


Figure 8: Received Eye Diagram after Maximum Energy Synchronization

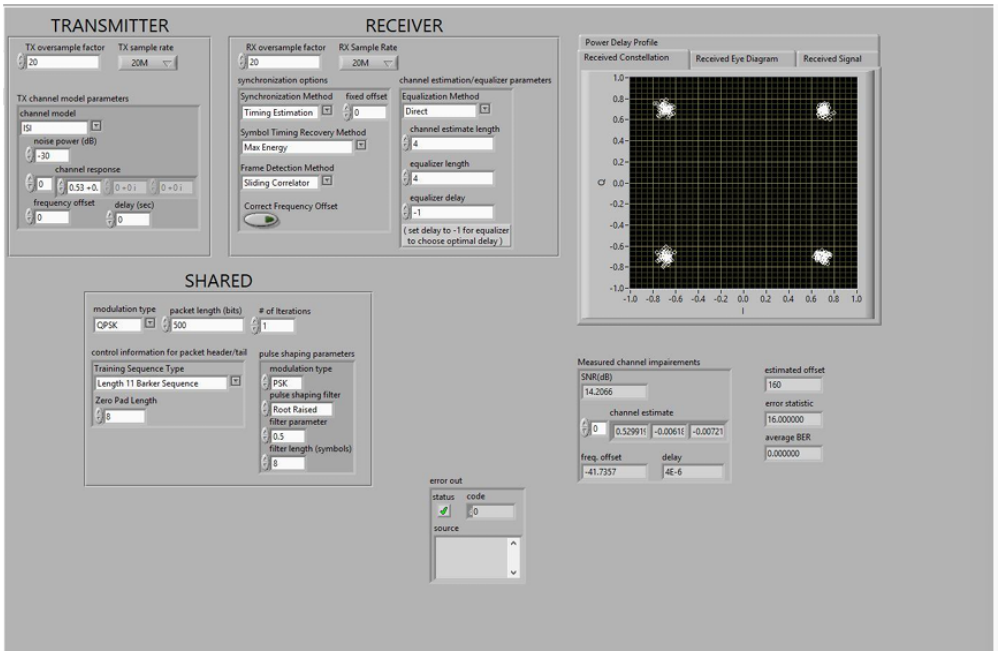


Figure 9: Received QPSK Constellation after Maximum Energy Synchronization

8 Results and Discussion

References

[1] H. Robert W, “Digital communications-Physical layer exploration lab using the NI USRP platform.”

Implementation of Channel Estimation and Equalization

Date Performed:
Expt. No.: 4
Reg. No.:
Name:

1 Objective

- To conduct an experiment and study the performance of channel estimation techniques with the aid of NI - USRP 2920/2901.
- To conduct an experiment and study the performance of equalization techniques with the aid of NI - USRP 2920/2901.

2 Equipments/Tool(s) Required

- NI - USRP 2920/2901
- VERT 900: 824-960 MHz, 1710-1990 MHz dual-band cellular and ISM vertical antenna
- LABVIEW

3 Theory

Channel Estimation

There are several different criteria for designing an estimator including the maximum likelihood criterion, minimum mean squared error, and least squares. Of these, perhaps the least squares technique is the simplest thus we will discuss it here. The added advantage of the least squares estimator is that in AWGN the least squares estimator is also the maximum likelihood estimator. Suppose that $\left\{t[n]\right\}_{n=0}^{N_t}$ is a known training sequence. This is a set of symbols that is known a priori. Suppose that the known training symbols are inserted into the transmitted sequence such that $s[n] = t[n]$ for $n = 1, 2, ..., N_t$.

To solve for the estimate of the channel using the LLSE solution, it is necessary to form a system of linear equations. Consider

$$y[n] = \sum_{l=0}^{L} h[l]s[n-l] + v[n]$$

where $s[n] = t[n], n = 1, 2, ..., N_t$. Since s[n] is unknown for $n \geq N_t$ (that is the unknown data), we need to write the squared error only in terms of the unknown data. With this in mind the least squares problem is to find the channel coefficients that minimize the squared error

$$\left\{\hat{h}[0], \hat{h}[1], ..., \hat{h}[L]\right\} = arg_{a[0],a[1],...,a[L]} min \sum_{n=L}^{N_t-1} ||y[n] - \sum_{l=0}^{L} a[l]t[n-l]||^2$$

The summation starts with n=L to ensure unknown data is not included. The least squares estimator is simply a generalization of the narrowband estimator. A "straightforward" way to solve this problem is to differentiate with respect to $a^*[m]$, build a set of linear equations, and solve. An alternative approach is to build a suitable set of linear equations. This is a powerful approach to solve a large class of least squares problems. First write the observed data as a function of unknown in matrix form

$$\begin{pmatrix} y[L] \\ y[L+1] \\ \vdots \\ y[N_t-1] \end{pmatrix} = \begin{pmatrix} t[L] & \dots & t[0] \\ t[L+1] & \ddots & \vdots \\ \vdots & & \vdots \\ t[N_t-1] & \dots & t[N_t-1-L] \end{pmatrix} \begin{pmatrix} a[0] \\ a[1] \\ \vdots \\ a[L] \end{pmatrix}$$

$$y = \mathbf{T}a$$

where we refer to **T** as the training matrix. If **T** is square or tall and full rank, then **T**$*$**T** is an invertible square matrix, where $*$ stands for conjugate transpose (Hermitian). The tall assumption (square with equality) requires that

$$\left\{\hat{h}[0], \hat{h}[1], ..., \hat{h}[L]\right\} = arg_{a[0],a[1],...,a[L]} min \sum_{n=L}^{N_t-1} ||y[n] - \sum_{l=0}^{L} a[l]t[n-l]||^2$$

$$N_t - L \geq L + 1$$

Or equivalently,

$$N_t \geq 2L + 1$$

Generally choosing N_t much larger than $L + 1$ (the length of the channel) gives better performance. The full rank condition can be guaranteed by ensuring that the training sequence is persistently exciting. Basically this means that it looks random enough. Random training sequences perform with good correlation properties generally satisfy this requirement.

Channel Equalizer Calculation:
With an estimate of the channel $\{t[l]\}_{l=0}^{L}$ in hand, the next task is to remove the effects of this channel.There are several approaches for designing a linear equalizer. In this lab we compute the least-squares equalizer. The objective is to find a filter such that

$$\sum_{n=1}^{\infty} f[l]\hat{h}[n-l] \approx \delta[n-n_d]$$

Except in trivial channels this equation cannot be satisfied exactly. The reason is that an FIR filter requires an IIR filter. The parameter n_d in this equation is the equalizer delay and is generally, a design parameter. Generally allowing $n_d > 0$ improves performance. The best equalizers consider several values of and choose the best one.
A straightforward approach is the least-squares equalizer. The key idea is to write a set of linear equations and solve for the filter coefficients that ensure that this equation minimizes the squared error. Writing this equation in matrix form

$$\begin{pmatrix} \hat{h}[0] & 0 & \dots & \dots \\ \hat{h}[1] & \hat{h}[0] & 0 & \dots \\ \vdots & \ddots & & \\ \hat{h}[L] & & & \\ 0 & \hat{h}[L] & \dots & \\ \vdots & & & \end{pmatrix} \begin{pmatrix} f[0] \\ f[1] \\ \vdots \\ \vdots \\ \vdots \\ f[L_f] \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ \vdots \\ 0 \end{pmatrix}$$

$$\hat{H}f = e_{n_d}$$

The matrix $\hat{H}$ is a type of Toeplitz matrix, sometimes called a filtering matrix. Assuming that $\hat{H}$ is full rank, which is guaranteed if any of the channel coefficients are nonzero, the linear least squares solution is

$$\hat{f}_{n_d} = (\hat{H} * \hat{H})^{-1}\hat{H} * e_{n_d}$$

The squared error is measured as $J_f[n_d] = ||\hat{H}\hat{f} - e_{n_d}||^2$. The squared error can be minimized further by choosing n_d such that $J_f[n_d]$ is minimized. This is known as optimizing the equalizer delay. The equalizer order is L_f. The choice of L_f is a design decision that depends on L. The parameter L is the extent of the multipath in the channeland is determined by the bandwidth of the signal as well as the maximum delay spread derived from propagation channel measurements. The equalizer is an FIR inverse of an FIR filter. As a consequence the results will improve if is L_f is large. The complexity required per symbol, however, also grows with L_f. Thus there is a tradeoff between choosing large L_f to have better equalizer performance and smaller L_f to have more efficient receiver implementation.

4 Flowchart

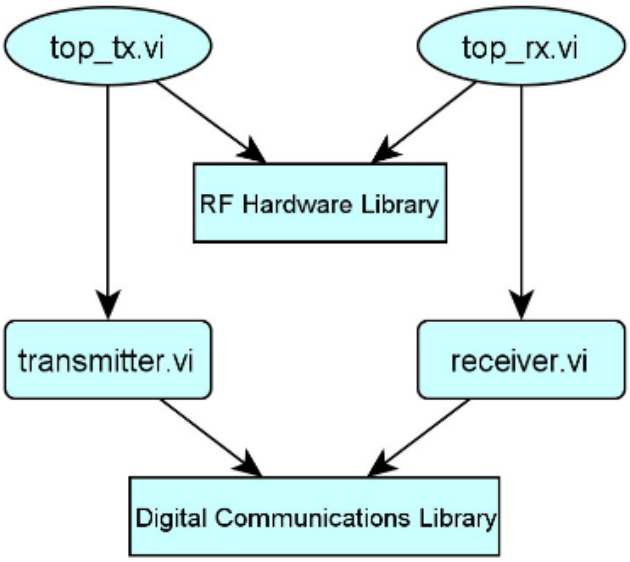


Figure 1: Hierarchy of code used in lab [1]

5 Block Diagram

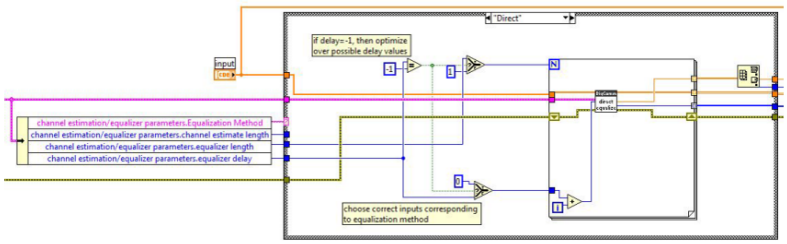


Figure 2: Equalizer.vi [1]

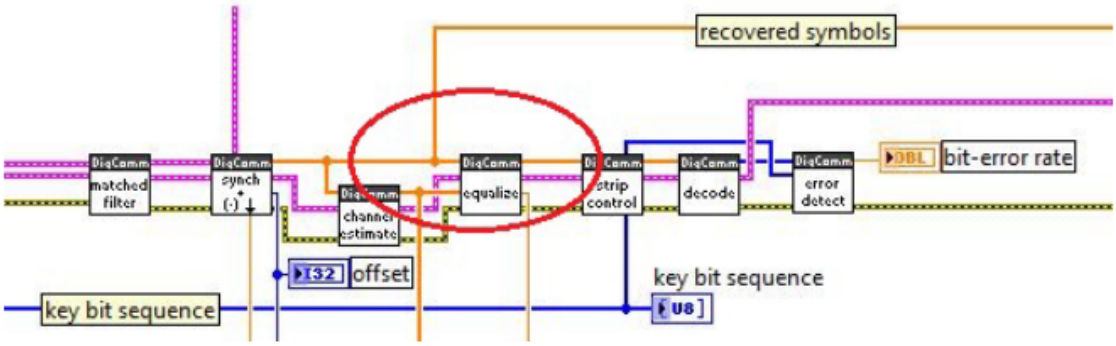


Figure 3: Modification of Receiver.vi needed to observer constellation prior to equalization. [1]

6 Procedure

Description of pulse LLSE.vi:

LLSE.vi – for linear system Ax = b, compute linear least squares estimate of parameter x			
INPUTS	A	2-D array of CDB (complex doubles)	m x n matrix
	b	1-D array of CDB	Length m vector
OUTPUTS	*x.estimate*	1-D array of CDB	Linear least squares estimates of parameter x
	Least squared error	DBL (double)	Squared error of estimate, $\|\|A\hat{x} - b\|\|^2$
	Error code	I32 (integer)	Error code generated by *Inverse Matrix.vi*

Figure 4: Modification of Receiver.vi needed to observer constellation prior to equalization. [1]

Description of pulse Toeplitz.vi:

Toeplitz.vi – generate an m-by-n Toeplitz structured matrix from the first row (length n) and column (length m) of the matrix			
INPUTS	*row*	1-D array of CDB (Complex doubles)	First row (length *n*) of the matrix
	column	1-D array of CDB	First column (length *m*) of the matrix
OUTPUTS	*Toeplitz array*	2-D array of CDB	*m*-by-*n* Toeplitz structured array

Figure 5: Modification of Receiver.vi needed to observer constellation prior to equalization. [1]

1.Considering a narrowband channel, setup the following parameters in your system using the appropriate controls in top_rx.vi and top_tx.vi

- Packet length = 500 bits
- Tx sample rate = 20M samples/sec
- Tx oversample factor = 200
- Rx sample rate = 2M samples/sec
- Equalizer length = 1
- Capture time = 3.5msec

2.Rewire a part of the block diagram, to observe the received constellation prior to equalization.
3.Considering wideband system, set the following parameters in your system.

- Packet length = 500 bits
- Tx sample rate = 20M samples/sec
- Tx oversample factor = 4
- Rx sample rate = 10M samples/sec
- Equalizer length = 6
- Rx oversample factor = 2
- Channel estimation length = 6
- Capture time = 80μ sec

4.Disable the frequency offset correction to avoid errors.
5.Also rewire the wire labeled ‘recovered symbols’.

7 Results and Conclusions

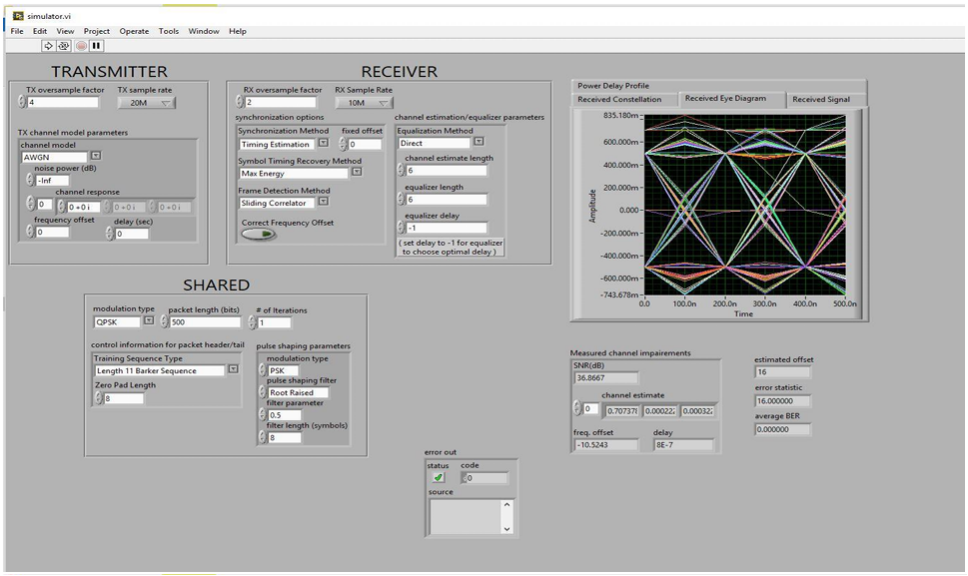


Figure 6: Received Eye Diagram for Narrowband ISI Channel

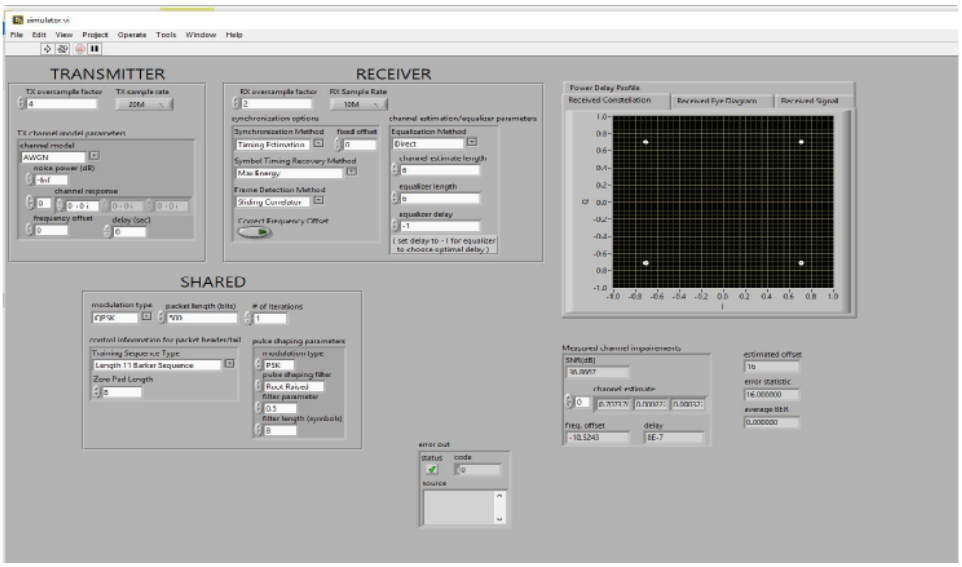


Figure 7: Received Eye Diagram for Wideband AWGN Channel

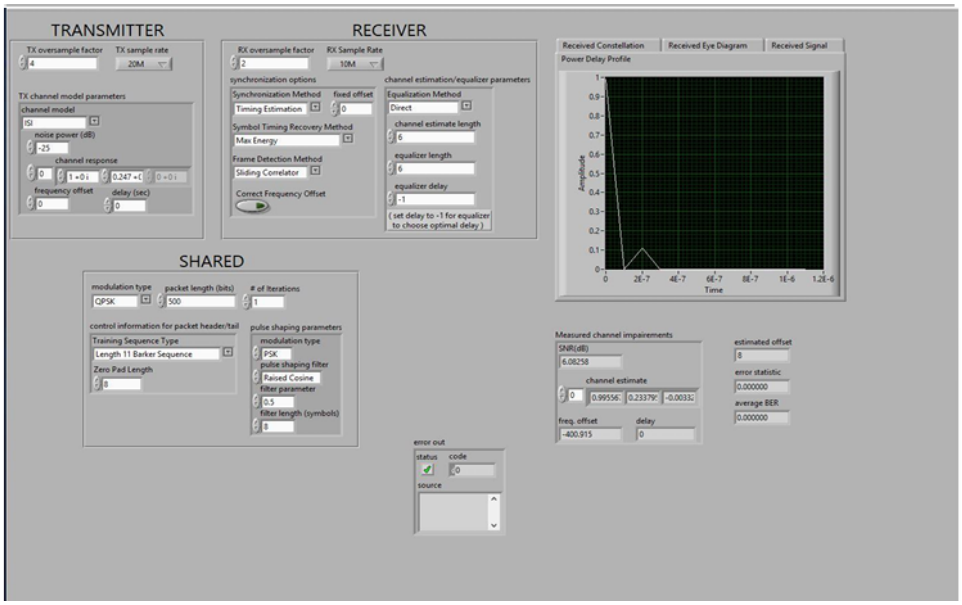


Figure 8: Received Constellation Diagram for Wideband AWGN Channel

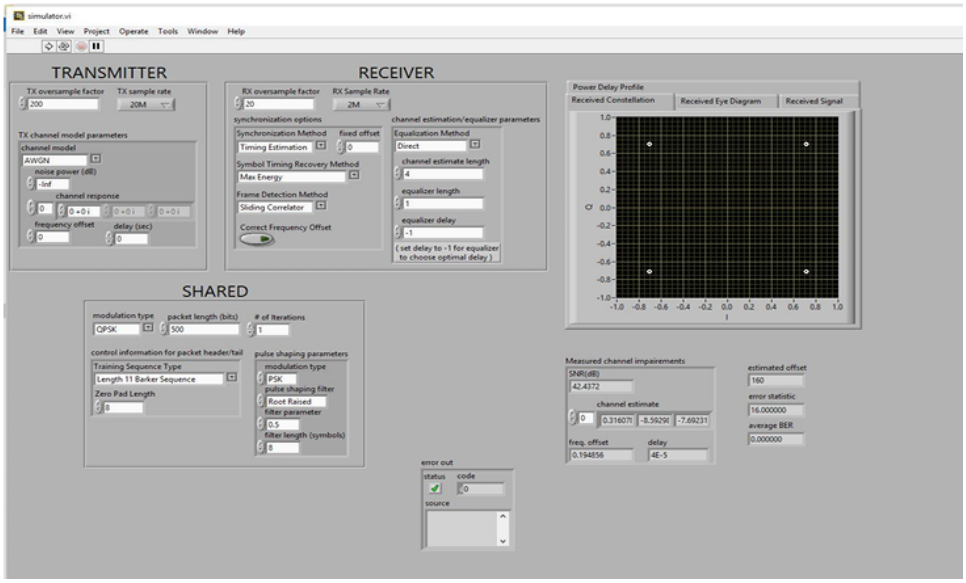


Figure 9: Power Delay Profile for Wideband ISI Channel

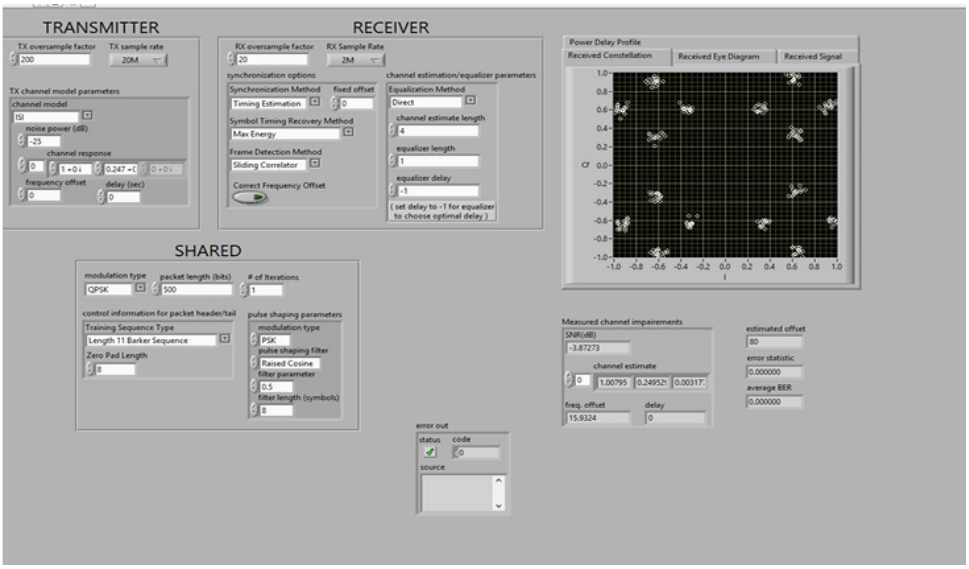


Figure 10: Received Constellation Diagram for Narrowband AWGN Channel

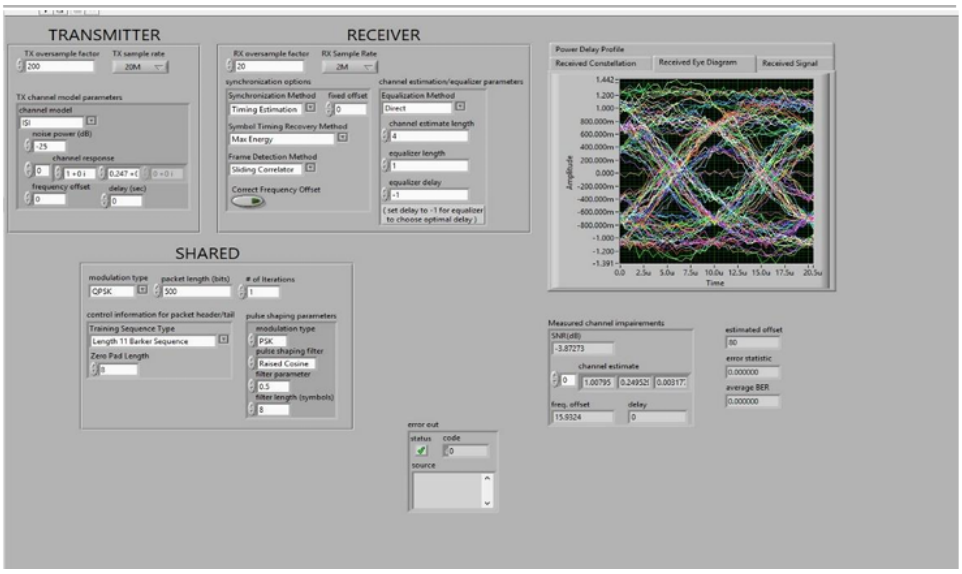


Figure 11: Received Constellation Diagram for Narrowband ISI Channel

8 Results and Discussion

References

[1] H. Robert W, “Digital communications-Physical layer exploration lab using the NI USRP platform.”

Implementation of Frame Detection and Frequency Offset Correction Techniques

Date Performed:
Expt. No.: 5
Reg. No.:
Name:

1 Objective

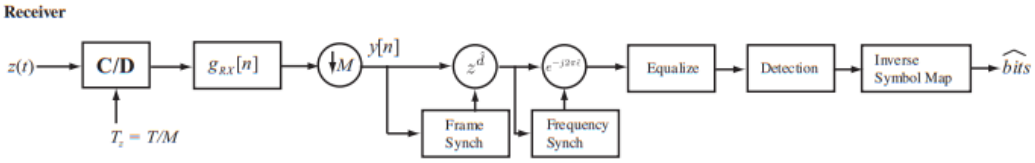
To conduct an experiment and study the concept of frame detection and frequency offset correction with the help of NI-USRP 2920/2901.

2 Equipments/Tool(s) Required

- NI-USRP 2920/2901
- VERT 900: 824-960 MHz, 1710-1990 MHz dual-band cellular and ISM vertical antenna
- LABVIEW

3 Theory

In practice, due to propagation and signal processing delays, the location of the beginning of the frame is unknown. This makes it difficult to apply the channel estimation and equalization algorithms. Up to this point it has also been assumed that the carrier frequency of the transmitter and receiver were locked together. Even small differences, however, lead to a distortion known as carrier frequency offset. So we will implement a particular method for frequency offset correction, called the Moose algorithm, and a method for frame detection based on correlation. The frequency offset correction and frame detection algorithms are used to correct these differences.



The new synchronization functionality is introduced after down sampling and before the equalization. The first step is determining the location of the training sequence, known as frame synchronization. The second step is estimation and correction of the carrier frequency offset, known as carrier frequency synchronization. The linear equalizer block includes the channel estimation and equalization functions.

The purpose of frame synchronization is to resolve multiple symbol period delays, determining what we call the beginning of the frame. Essentially the objective is to find a known reference in the transmitted signal so that subsequent receiver operations can proceed like channel estimation and equalization.

Passband communication signals are used in wireless communication systems. A passband signal has energy that is nonzero for a frequency band concentrated about some carrier frequency fc. At the transmitter the baseband signal is upconverted to a passband signal while at the receiver a passband signal is down-converter to a baseband signal. In most systems the carrier frequency used for up conversion at the transmitter does not match the carrier frequency at the receiver, as illustrated in Figure 2. This creates what is known as a carrier frequency offset.

4 Block Diagram

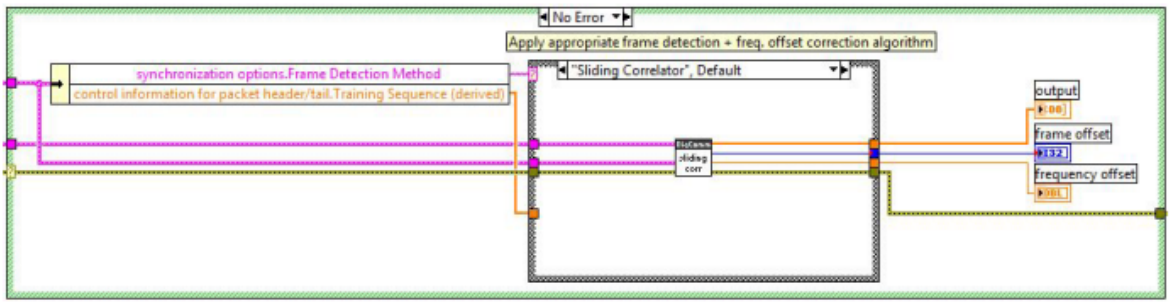


Figure 1: Framedetect.vi [1]

5 Procedure

Moose.vi - implements the Moose algorithm			
Inputs	input	1-D array of CDB (complex doubles)	received sequence after frame detection
	symbol rate	DBL (double)	symbol rate of the system
	channel estimate length	I32 (integer)	$L_h + 1$
Outputs	output	1-D array of CDB	received sequence after correcting for estimated frequency offset
	frequency offset	DBL	frequency offset estimate $\hat{f}_{offset} = \hat{\epsilon}/T$

In this lab you will run your implementation of sliding correlator.vi over a real wireless link. Using your code from the prior labs in conjunction with the new blocks you have created, you will complete the framework for the transmitter and receiver blocks in lab as in previous labs. Insert your code into this framework as you did in the pre-lab. Next, set up the following parameters in your system.

- TX sample rate = 4 MSamp/sec
- TX oversample factor = 20
- RX sample rate = 4 MSamp/sec
- RX oversample factor = 20
- Channel estimate length = 2
- Equalizer length = 4
- Capture time = 2 msec

Leave any unspecified parameters set to their default values. Also make sure that the *Correct Frequency Offset* control on top_rx is set to TRUE. You will now perform two experiments to explore frequency offsets and frame detection over a real wireless link. Before you begin, estimate the order of magnitude of the frequency offset between the transmitter and receiver of your system. Take note of this value as you may be asked about it later.

6 Results and Conclusions

a. Frame Offset = 0

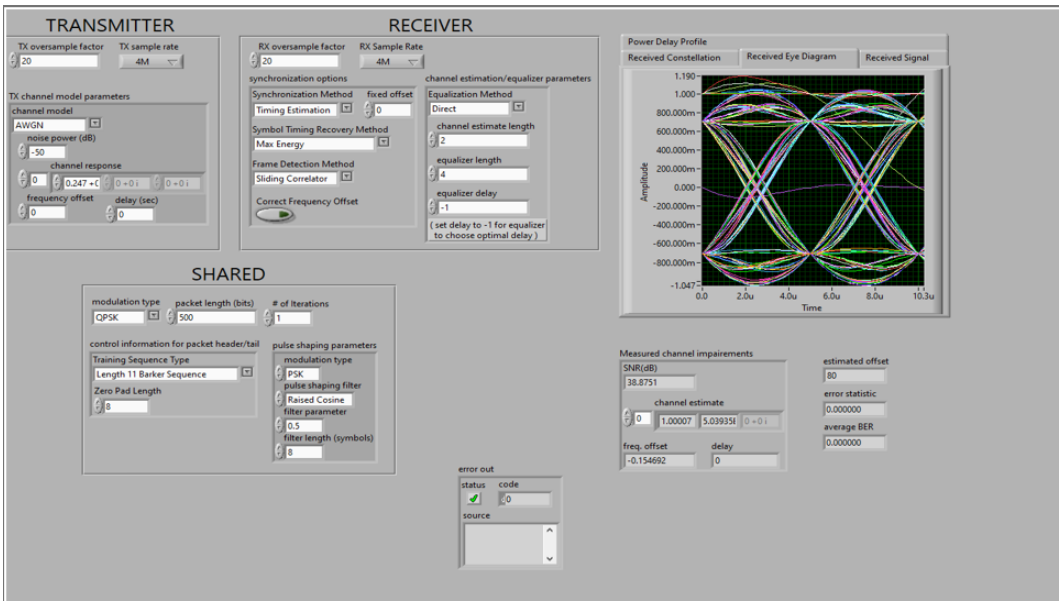


Figure 2: Eye diagram

b. Frame Offset = 30

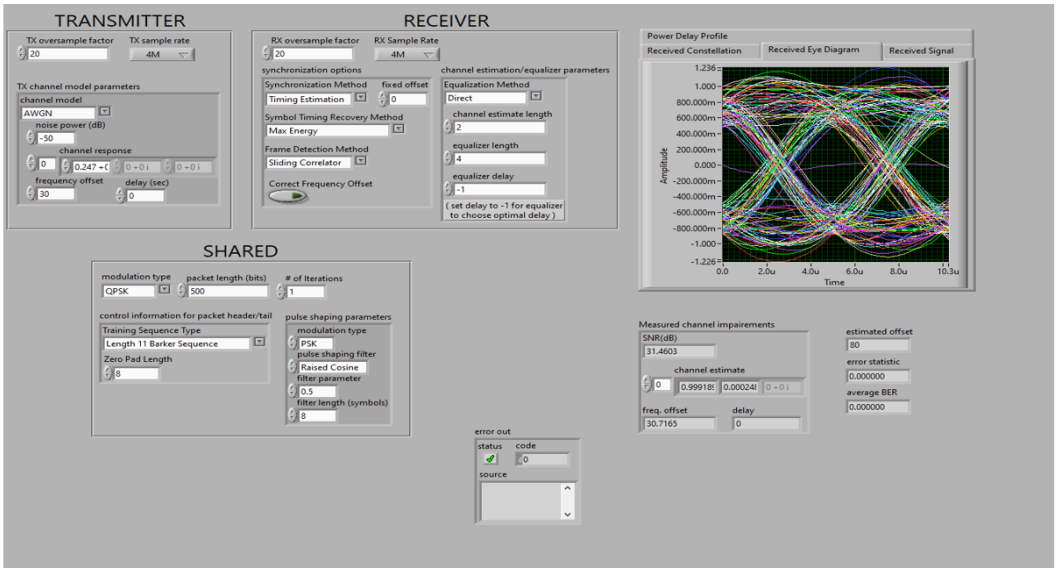


Figure 3: Eye diagram

c. Frame Offset = 60

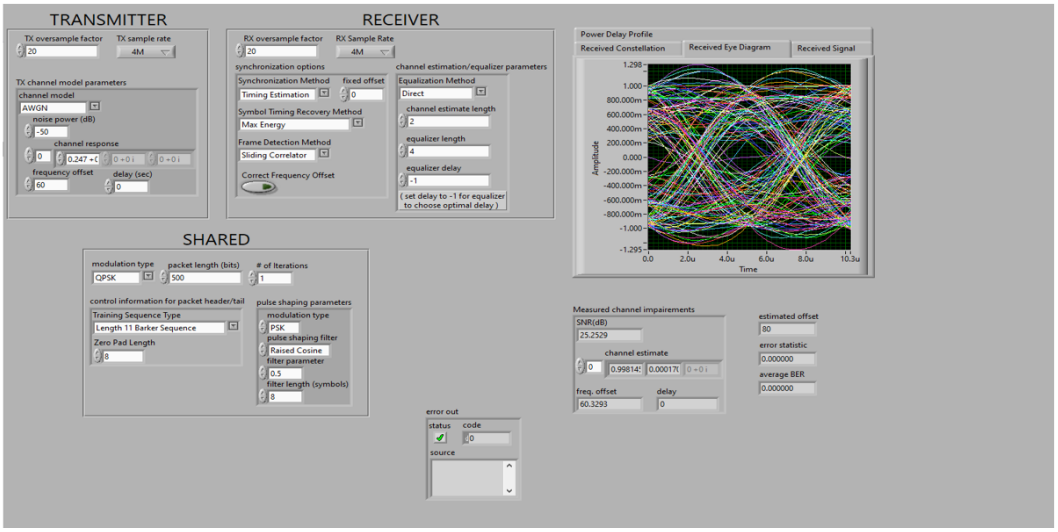


Figure 4: Eye diagram

d. Frame Offset = 100

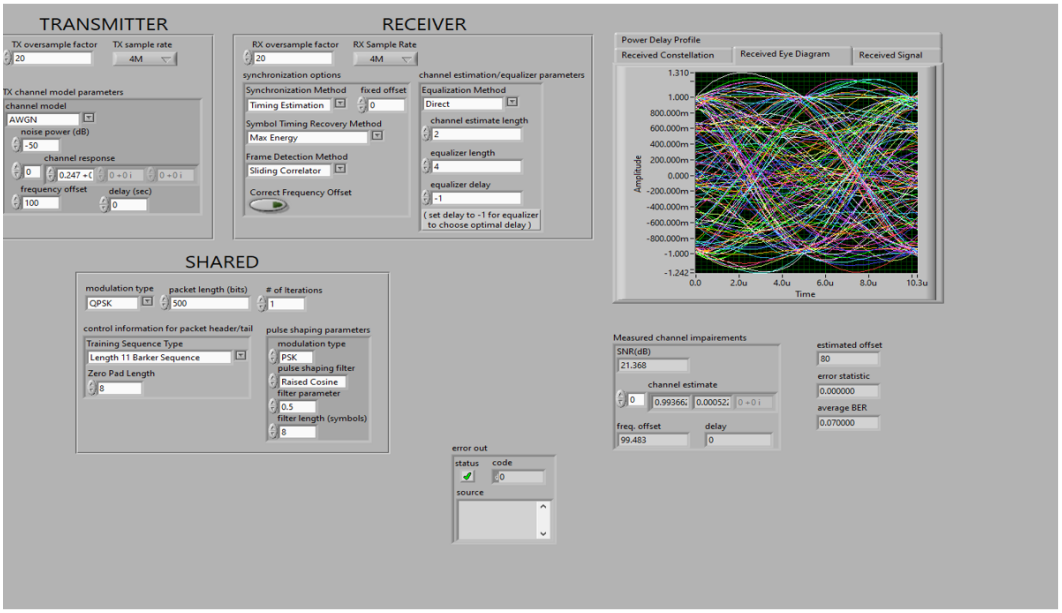


Figure 5: Eye diagram

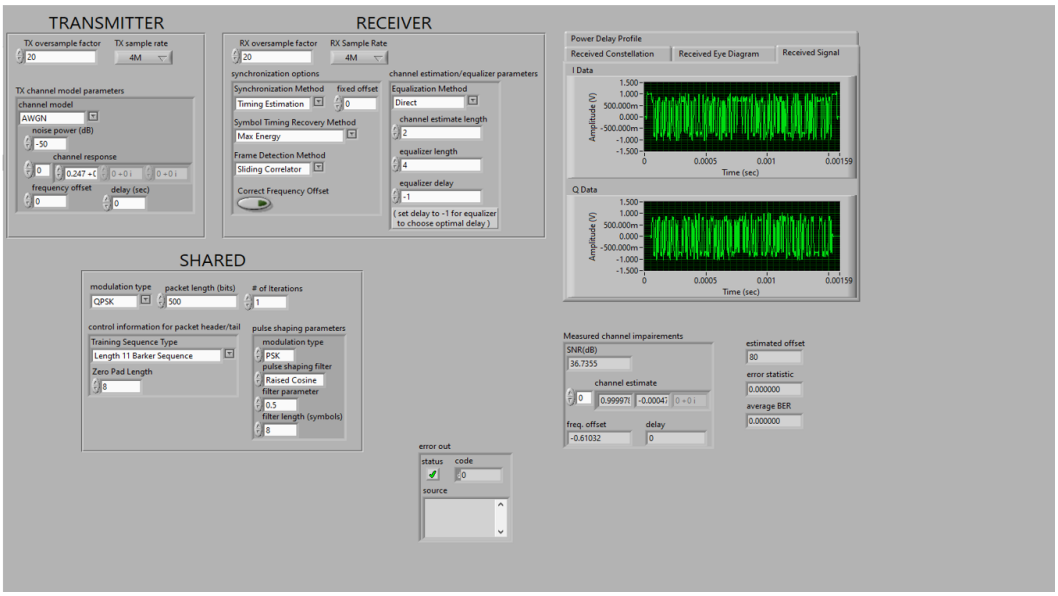


Figure 6: Received Signal

7 Results and Discussion

References

[1] H. Robert W, “Digital communications-Physical layer exploration lab using the NI USRP platform.”

Implementation of OFDM Modulation

Date Performed:
Expt. No.: 6
Reg. No.:
Name:

1 Objective

To conduct an experiment and study the performance of OFDM with the aid of NI - USRP 2920/2901.

2 Equipments/Tool(s) Required

- NI - USRP 2920/2901
- VERT 900: 824-960 MHz, 1710-1990 MHz dual-band cellular and ISM vertical antenna
- LABVIEW

3 Theory

OFDM is a type of digital modulation where information is modulated into discrete-time sinusoids. With OFDM, the symbols after the constellation mapping are considered to start in the frequency domain. OFDM operates on groups of symbols, the resulting group of symbols being called an OFDM symbol. To begin the explanation, we assume *K = 0* and that there are *N* frequency-domain symbols in one OFDM symbol. The number of subcarriers is given by *N*, typically a power of 2. The transmitter operates as follows. Given $s[n]_{n=0}^{N-1}$ and cyclic prefix of length L_c, the transmitter produces the sequence

$$\frac{1}{N}\sum_{n=0}^{N-1} s[n]e^{\frac{j2\pi(n-L_c)}{N}}$$

where *n = 0, 1, … N + L_c – 1*

that is passed to the transmit pulse shaping filter. The samples from *n = L_c, $L_c + 1$,...* , $N + L_c 1$ are the output of the inverse discrete Fourier transform (DFT) of the input symbols $s[n]_{n=0}^{N-1}$, which could be implemented using the fast Fourier transform (FFT).
Now observe that,

$$w[n] = w[n + N]$$

for *n = 0, 1, ...,* $L_C - 1$. This means that the first L_c samples of $w[n]$ are the same as the last L_c samples. The first L_c samples are known as a cyclic prefix. The length of the cyclic prefix L_c should be at least as long as L_h. This will ensure that there is no inter-symbol interference between adjacent OFDM symbols (i.e., the CP "guards" against ISI). The cyclic prefix serves another important purpose. It helps convert (part of) the linear convolution into a circular convolution. In OFDM rectangular pulse-shaping is common; windowing of the symbols is used to shape the frequency spectrum. This is not considered in this lab. Assuming carrier frequency synchronization and frame synchronization have been accomplished, the received observes the following signal after matched filtering, symbol timing, frame synchronization, and downsampling

$$y[n] = \sum_{b=0}^{L} h[l]w[n - l] + v[n]$$

The receiver operates on blocks of data.
Assuming that the indexing starts at the first OFDM symbol, the receiver discards the first L_c samples to form for *n = 0, 1, ..., N 1* (neglecting noise) $\overline{y[n]} = y[n + Lc]$. With some mathematical manipulations it can be shown that

$$\overline{y[n]} = \sum_{h=0}^{L} h[l]w[n + L_c - l]$$

$$= \frac{1}{N} \sum_{m=0}^{N-1} (\sum_{l=0}^{L} h[l]e^{-j2\pi \frac{ml}{N}}) s[m]e^{j2\pi \frac{mn}{N}}$$

n = 0, 1, ..., N - 1

The receiver then takes the DFT (or FFT) of these samples to produce (after some simplification)

$$\overline{Y[k]} = DFT[\overline{y[n]} = H[k]s[k] + V[k] \tag{1}$$

where

$$H[k] = \sum_{l=0}^{L} h[l]e^{-j2\pi \frac{kl}{N}}$$

which is just the DFT of the zero padded channel. The frequency domain interpretation of OFDM comes from Eq. (1). Essentially information is sent on discrete-time sinusoids, or carriers. The information on the kth discrete-time sinusoid experiences the channel response determined by $H[k]$. Equalization simply requires dividing $\overline{Y[k]}$ by $H[k]$.

4 Flowchart

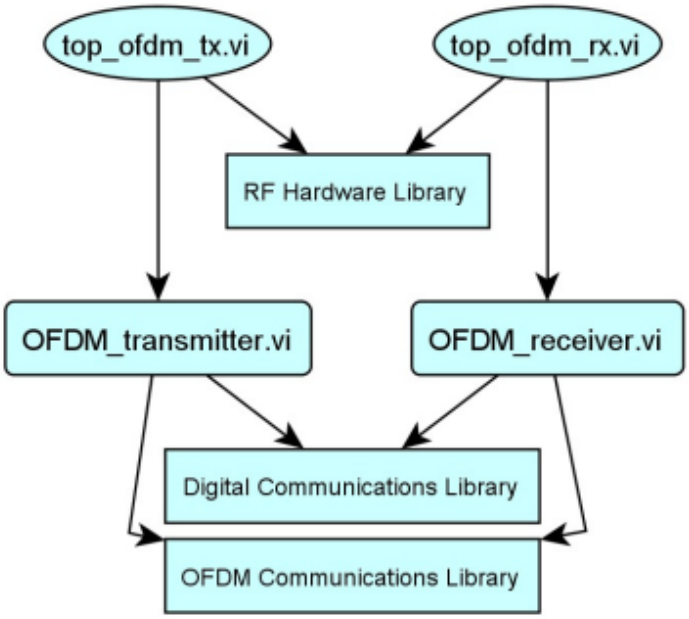


Figure 1: Hierarchy of files for OFDM framework [1]

5 Block Diagram

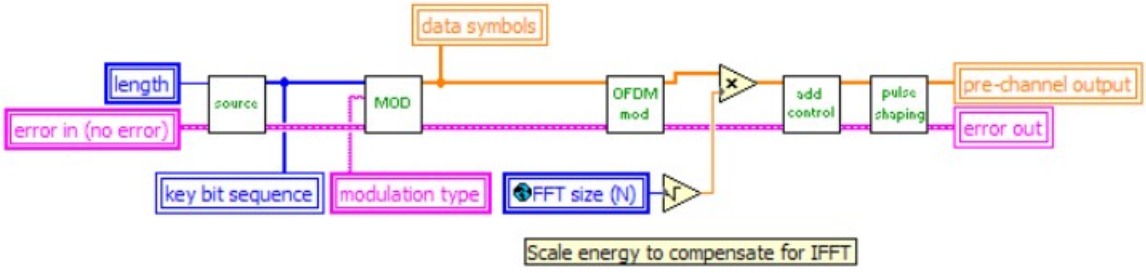


Figure 2: Transmitter.vi [1]

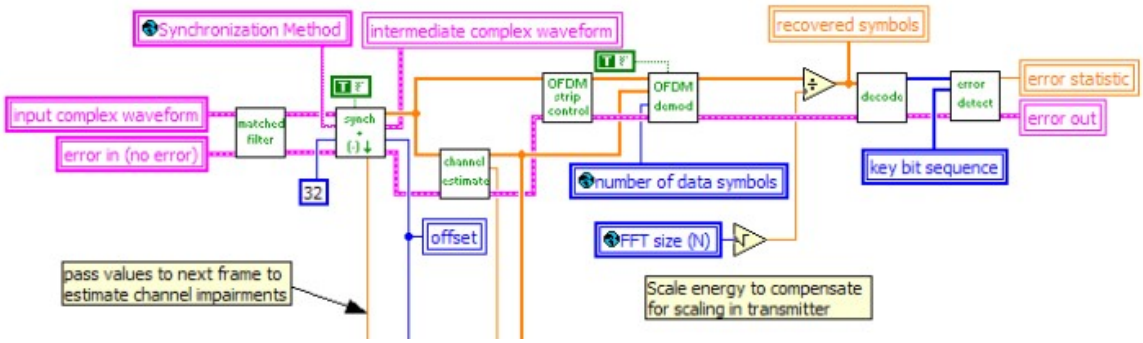


Figure 3: Receiver.vi [1]

6 Procedure

a. Construct modulate.vi and demodulate.vi files with the following parameters.

OFDM modulate.vi-implements the OFDM modulator			
INPUTS	*inputsymbols*	1 - Darray of CDB (Complex doubles)	input symbol stream to be modulated using OFDM
OUTPUTS	*outputsamples*	1 - Darray of CDB	output sample stream after OFDM modulation; one OFDM symbol corresponds to *(N+Lc)* samples

b. Construct top_tx.vi, top_rx.vi with the following parameters

- Packet length = 500 bits
- Modulation type = QPSK
- Channel estimate length = 4
- FFT size (N) = 64
- Length of CP *(Lc)* = 8
- Null tones = {0, 31, 32, 33}

c. Setup the following parameters for a narrowband system

- TX sample rate = 4 MSamp/sec
- TX oversample factor = 20
- RX sample rate = 4 MSamp/sec
- RX oversample factor = 20

OFDM demodulate.vi - implements the OFDM demodulator			
INPUTS	*inputsamples*	1-Darray of CDB	input sample stream to be demodulated using OFDM; one OFDM symbol corresponds to *(N + Lc)* samples
	channelestimate	1-Darray of CDB	channel estimate computed prior to demodulation (used for FEQ)
	numberof datasymbols	I32	number of data symbols to be recovered
	equalizechannel?	Boolean	determines whether or not you should apply FEQ to OFDM demodulated symbols; set this value to TRUE by default
OUTPUTS	textitdemodulatedsymbols	1-Darray of CDB	stream of output symbols after OFDM modulation and FEQ (when appropriate)
	FD channelestimate	1-Darray of CDB	frequency domain response of channel estimate computed taking DFT of channel estimate

- Capture time = 2.4 msec

After transmitting a packet successfully, observe the frequency response of the narrowband channel, using the Channel Response graph located on the front panel of OFDM receiver.vi. Also examine the instantaneous power delay profile using the appropriate graph on the front panel of the same VI.

d. To setup a wideband system, set the following parameters in your OFDM system

- TX sample rate = 20 MSamp/sec
- TX oversample factor = 4
- RX sample rate = 10 MSamp/sec
- RX oversample factor = 2
- Capture time = 100 μsec

After transmitting a packet successfully observe the frequency response and power-delay profile of the wideband channel in OFDM receiver.vi.

e. To observe how the performance of an OFDM system degrades in the presence of a frequency offset. Set up the following parameters in your system

   - TX sample rate = 20 MSamp/sec
   - TX oversample factor = 20
   - RX sample rate = 4 MSamp/sec
   - RX oversample factor = 4
   - Capture time = 500 μsec
   - Frequency offset (Hz) = 200Hz

f. Set Correct Frequency Offset. Now consider two OFDM systems which use different values for *N*. First, consider a system which uses *N=64* subcarriers. Set up the following OFDM parameters

   - FFT Size (*N*) = 64
   - Length of CP (L_C) = 8
   - Null tones = $\{0, 31, 32, 33\}$

g. Observe how a frequency offset impacts the received signal constellation. Also make note of the impact of a frequency offset on BER performance. Increase the amount of frequency offset in your system in increments of 200Hz and observe how the signal constellation and BER performance vary. Now consider a system which uses N=1024 subcarriers

   - FFT Size (*N*) = 1024
   - Length of CP L_C = 32
   - Null tones = $\{0, 511, 512, 513\}$

7 Results and Conclusions

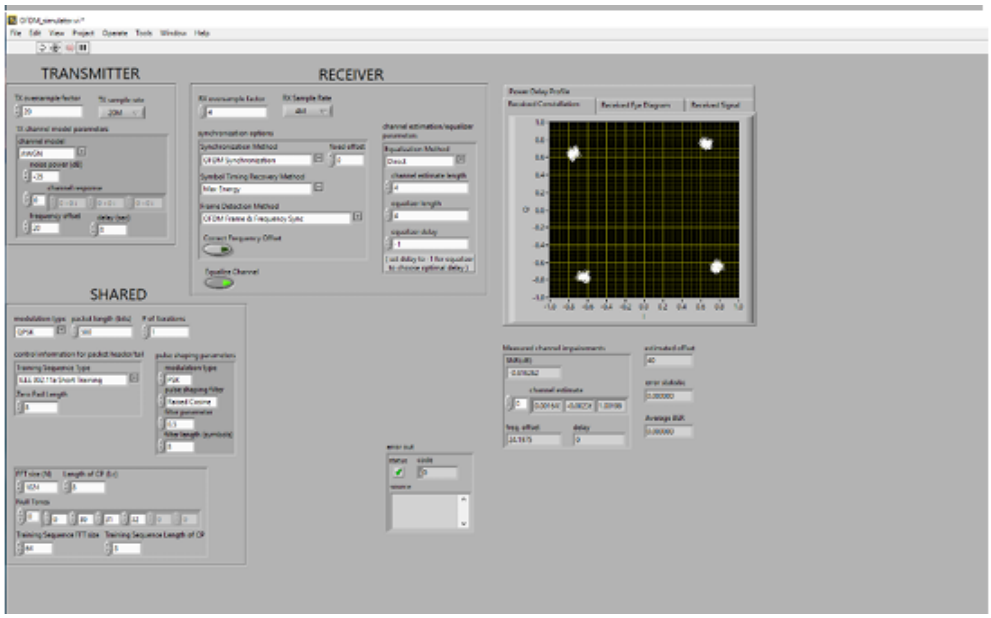


Figure 4: Received Constellation for Narrowband OFDM System

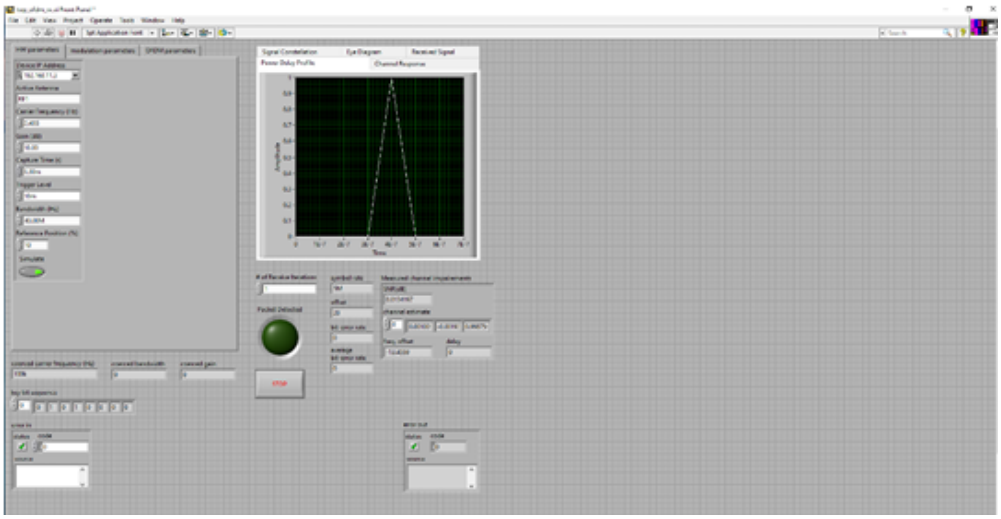


Figure 5: Received Constellation for Wideband OFDM System

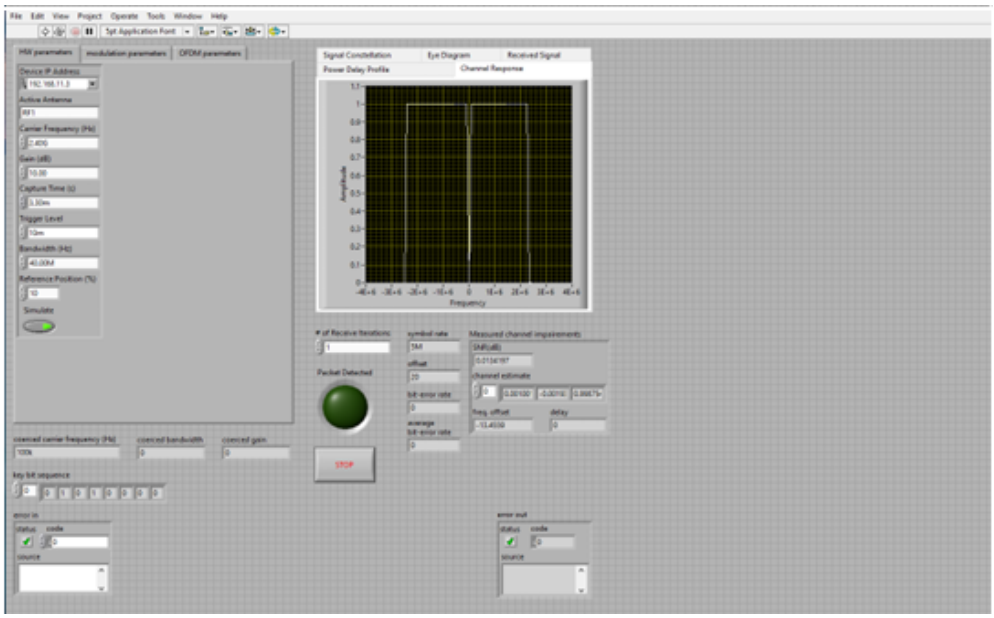


Figure 6: Channel Response of the OFDM System

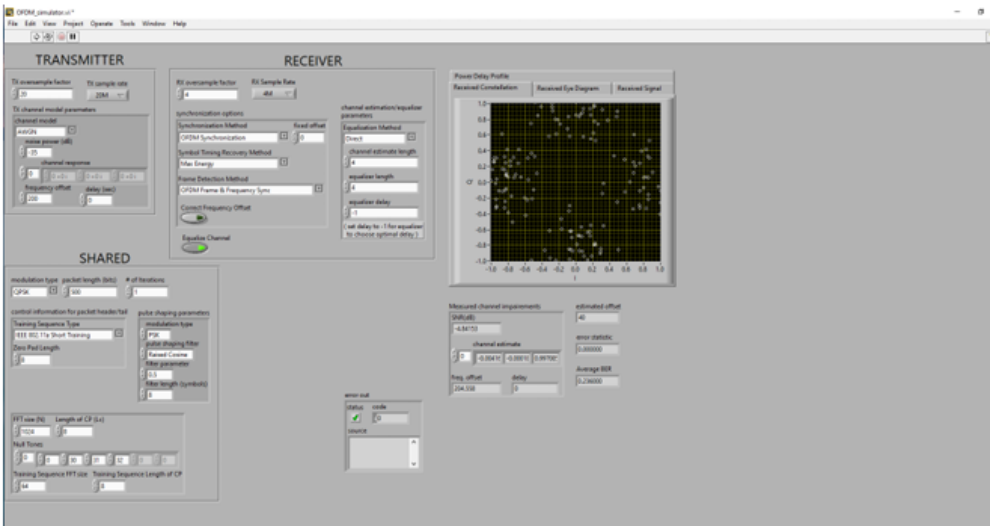


Figure 7: Effect of Frequency Offset on the Received OFDM Constellation

8 Results and Discussion

References

[1] H. Robert W, “Digital communications-Physical layer exploration lab using the NI USRP platform.”

Channel Coding in OFDM Systems

Date Performed:
Expt. No.: 7
Reg. No.:
Name:

1 Objective

To conduct an experiment and study the concept of channel coding in OFDM systems with the aid of NI - USRP 2920/2901.

2 Equipments/Tool(s) Required

- NI - USRP 2920/2901
- VERT 900: 824-960 MHz, 1710-1990 MHz dual-band cellular and ISM vertical antenna
- LABVIEW

3 Theory

Coding theory is a field which encompasses both source and channel coding. It is largely based on results from finite field algebra. The main idea behind channel coding is that adding structured redundancy to the source bits for a transmitted sequence can improve the resilience of transmission to channel distortion. In other words, a properly designed channel code can improve the bit-error rate (BER) performance of a system (i.e., reducing the transmit power required to achieve a particular error performance). A rate $\frac{k}{n}$ code takes a length L bit sequence and generates $\frac{nL}{k}$ bits.
There are two main classes of channel codes: (1) block codes and (2) convolutional codes. Channel coding is sometimes referred to as forward error correction (FEC) or error control coding, or just “coding”.

CHANNEL CODE	CODE TYPE	TECHNOLOGY/STANDARD
Hamming Code	Block	-
Reed–Solomon	Block	disk drives, CD's, DVD's, WLAN
Punctured Codes	Convolutional	Hybrid-ARQ, IEEE 802.11a
Turbo Codes	Convolutional	3G mobile, satellite comm.
Low Density Parity Check (LDPC)	Block	-

3.1 Block Codes

The encoder first segments the source sequence into blocks of k bits. Thus each block represents any one of $2k$ distinct messages. This block is in turn mapped to a length n bit sequence called a codeword. Each codeword contains $r = n - k$ redundant bits. This kind of code is called an (n, k) block code and can be used to detect and possibly correct errors. The resilience of block codes to channel distortion largely depends on the structure of the r redundant bits.

3.2 Convolution Codes

A class of channel codes which employ convolutional encoders. The encoded sequence is a convolution of the algebraic polynomial describing the source bit sequence with a polynomial which describes the encoder. Equivalently, the ith codeword generated by a convolutional encoder is a weighted sum of the previous input messages to the encoder. This is similar to the convolution used in linear time-invariant systems and shares many of the same properties. Convolutional codes generally have very efficient implementations which employ shift registers (to retain state or memory) and exclusive-or (XOR) gates. Block codes, which generally require more logic to implement, are much easier to analyse. There are a number of trade-offs between both classes of codes, but neither class of codes fundamentally provides better error protection than the other

4 Flowchart

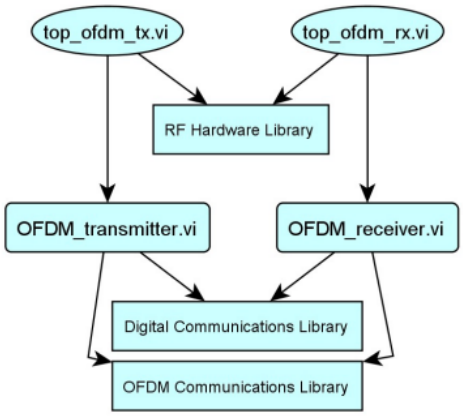


Figure 1: Hierarchy of files for OFDM framework [1]

5 Block Diagram

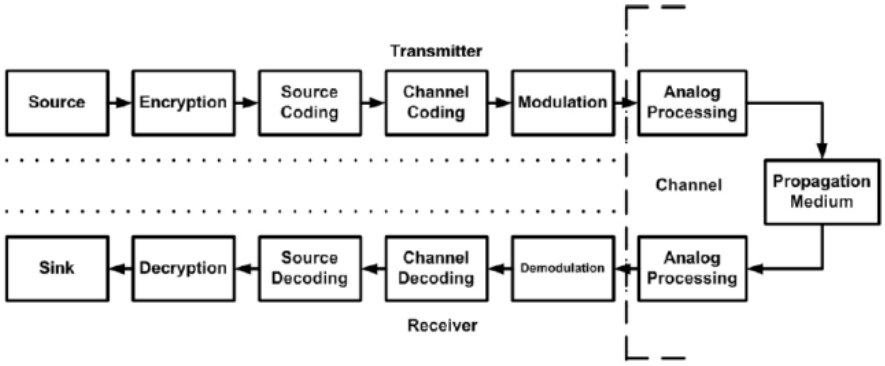


Figure 2: Transmitter and receiver block diagram [1]

6 Procedure

repeat_encoder.vi - implements rate (1/3) repetition code			
INPUTS	information bits sequence	1-D array of U8	bit sequence to be repetition encoded
	enable channel coding	Boolean	determines whether or not your VI should be using repetition code
OUTPUTS	encoded bit sequence	1-D array of U8	bit sequence after repetition encoding

repeat_decoder.vi - implements majority decoding to decode rate (1/3) repetition code			
INPUTS	estimated bit sequence	1-D array of U8	estimated bit sequence (i.e., after demodulation)
	enable channel coding	Boolean	determines whether or not your VI should be using repetition code
OUTPUTS	encoded bit sequence	1-D array of U8	bit sequence after majority decoding

7 Results and Conclusions

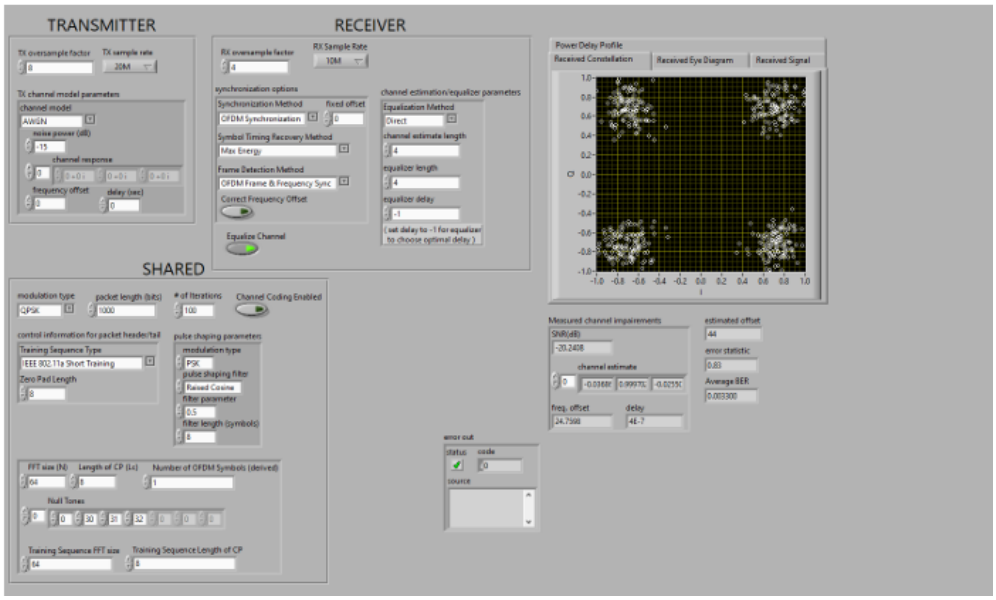


Figure 3: Received Constellation Diagram without Channel Coding

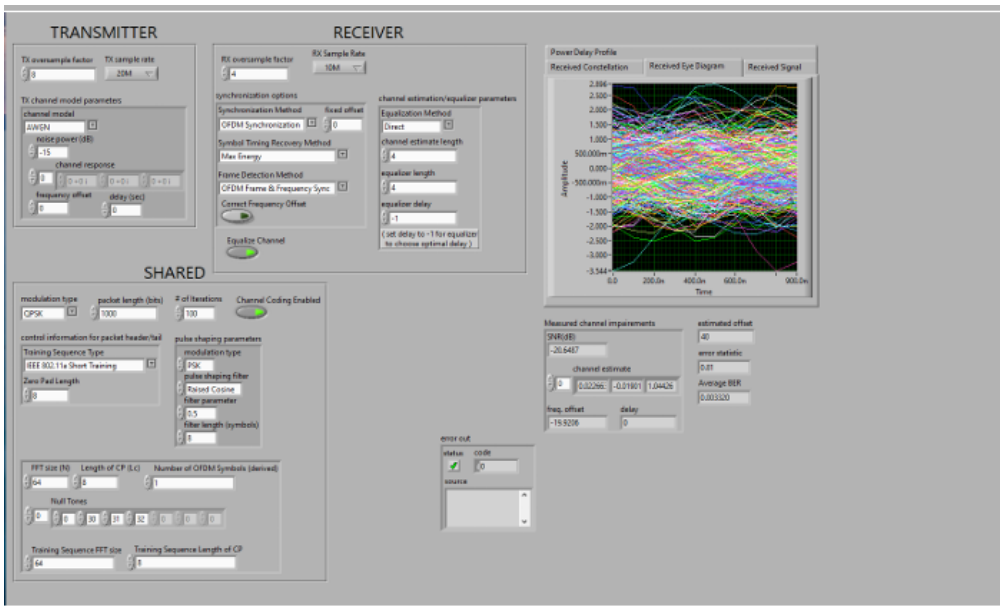


Figure 4: Received Eye Diagram without Channel Coding

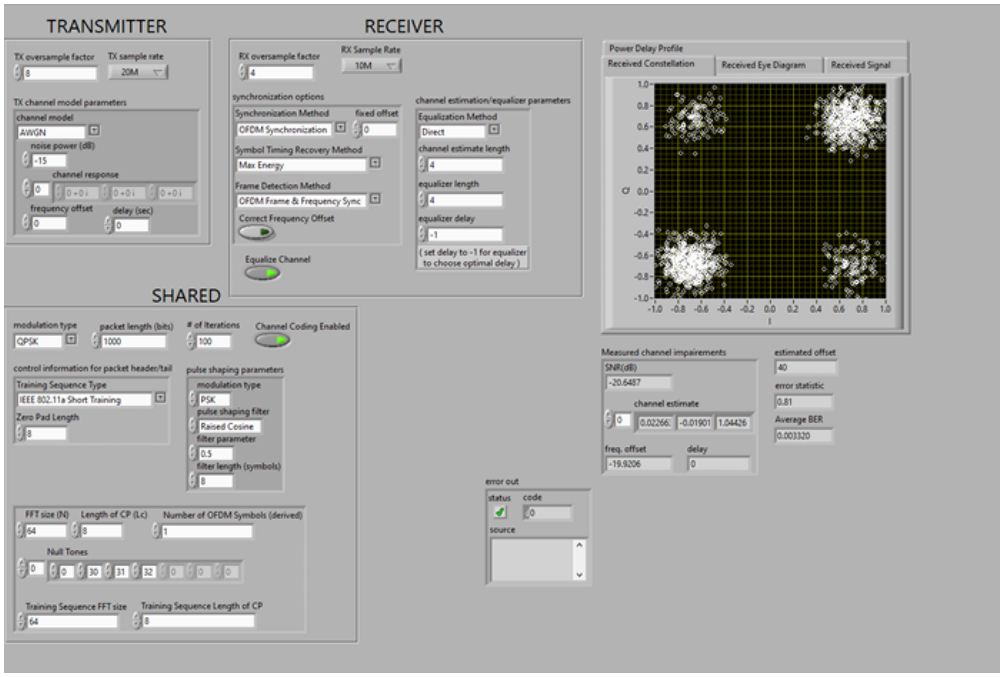


Figure 5: Received Constellation Diagram with Channel Coding

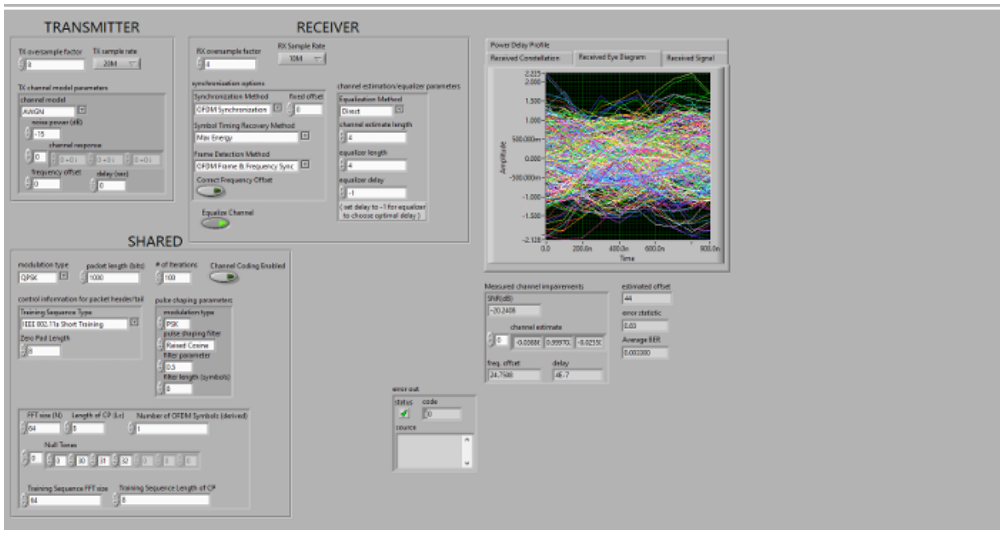


Figure 6: Received Eye Diagram with Channel Coding

8 Results and Discussion

References

[1] H. Robert W, “Digital communications-Physical layer exploration lab using the NI USRP platform.”

Expt. No. :

Date :

MEASUREMENT OF CONNECTOR AND BENDING LOSS OF OPTICAL FIBER

OBJECTIVE:

To measure the bending loss and coupling loss in an optic fiber communication link.

EQUIPMENT REQUIRED:

- Optical Fiber Testing Kit
- Function Generator
- CRO
- Probe
- Jump Wires
- Fiber Alignment Kit

THEORY:

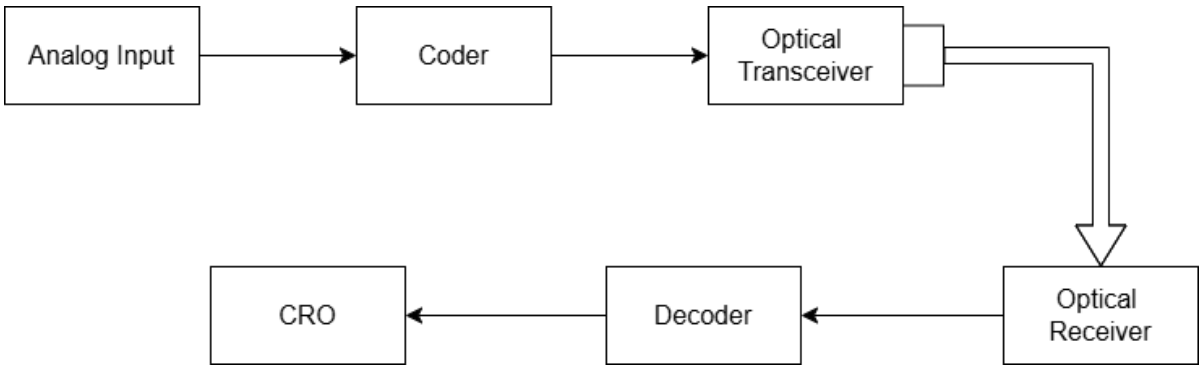
Attenuation to fiber optics is also known as transmission loss. Fiber can be subjected to two types of bends:

(a) Microscopic bends that have radii larger then when compared with fiber diameter.
(b) Random microscopic of fiber optic axis that can arise when the optic fiberare incorporated into cables

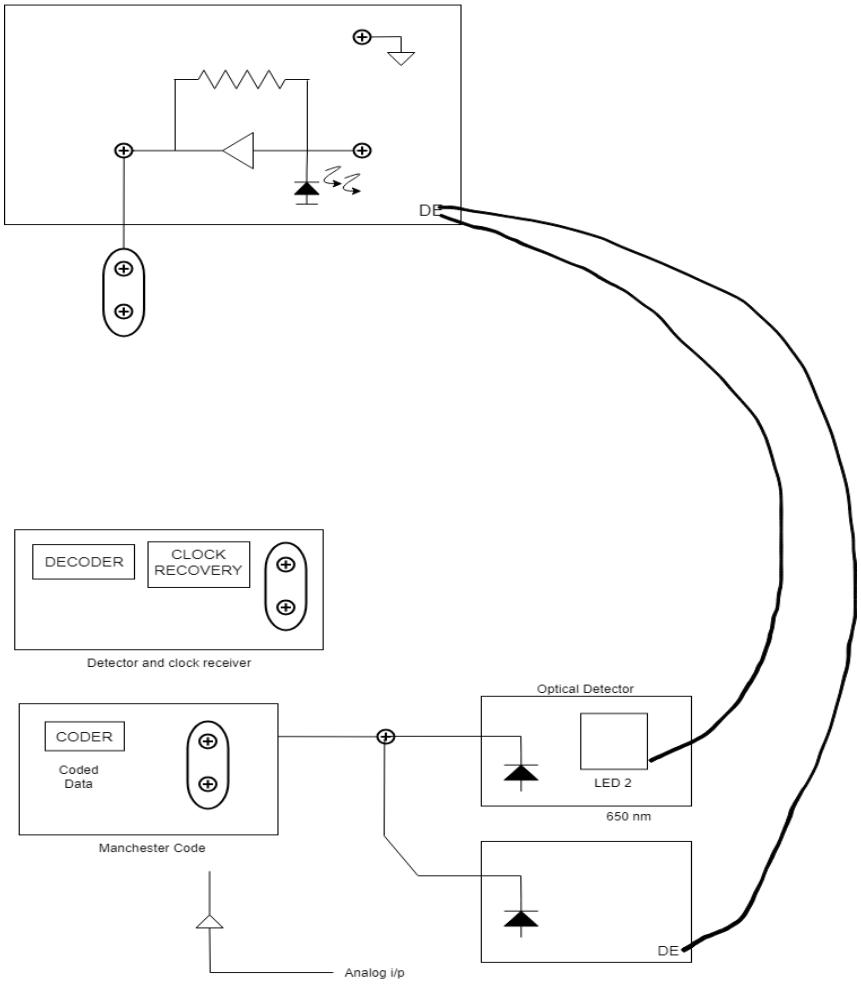
Macrobending is simply called as bending loss. For slight bends that loss is extremely small and is unobservable. If the bend radius is made a bit smaller once the threshold point has been reached the losses become suddenly extremely large.

The amount of optical radiation from a bent fiber depends on the field strength of X_c (critical distance) and on the radius of curvature 'R'.

BLOCK DIAGRAM:



KIT DIAGRAM:



PROCEDURE:

1. Set up the 850nm analog line using 3m fiber.

2. Derive a 1V p-p sinusoidal signal of 10 KHz with 0 DC at P_{11} and observe the received signal at P_{DC} on the oscilloscope.

3. Now load the fiber in a loop.

4. Reduce the diameter of the loop slowly and observe the reduction of the received signal.

5. Keep reducing the diameter of the loop to about 2 cm and plot the amplitude vs diameter graph

6. Connect one end of 1mfiber cable tofiber alignment cable and its other end to FDI, another 3m fiber cable is connected from LED2 to another end of fiber alignment kit.

7. Measure the received signals voltage and now keep increasing the distance between the two ends of wires in fiber alignment kit and observe the received signal voltage on CRO. 8. Plot the distance versus amplitude graph

OBSERVATION:

Bending Loss:

f = 1KHz ; Input Voltage V_0 = 1.07 V

Diameter (cm)	Amplitude of Output Voltage V_{out} (V)

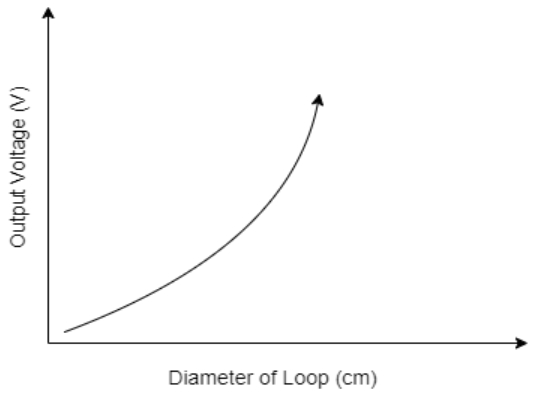
Connector Loss:

f = 1KHz ; Input Voltage V_0 = 1.07 V

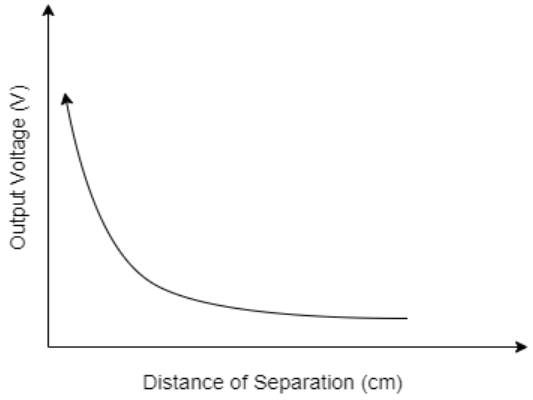
Separation Distance (cm)	Amplitude of Output Voltage V_{out} (mV)

MODEL GRAPHS:

Bending Loss:



Connector Loss:



RESULTS AND DISCUSSION:

Expt. No. :

Date :

ATTENUATION MEASUREMENT OF FIBER

OBJECTIVE:

To measure attenuation loss in the given optical fiber.

SOFTWARE USED:

- Optical Fiber
- Two Channe 20 MHz Oscillator
- Fiber Alignment Unit

THEORY:

Attenuation of light signals as it propagates along afiber is an important consideration in the design of an optical communication system.It plays an important role in determining the maximum distance between the transmitter and receiver.

The basic attenuator mechanism in the fiberare absorption scattering and radiative losses in the optical energy. Absorption is related to fiber material and structural imperfection in the optical waveguide.

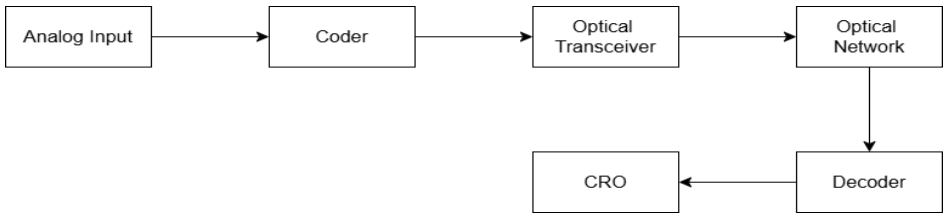
This attenuation increases length of running cable increases.

Other important losses occurring in the fibre are bending loss and connector loss.

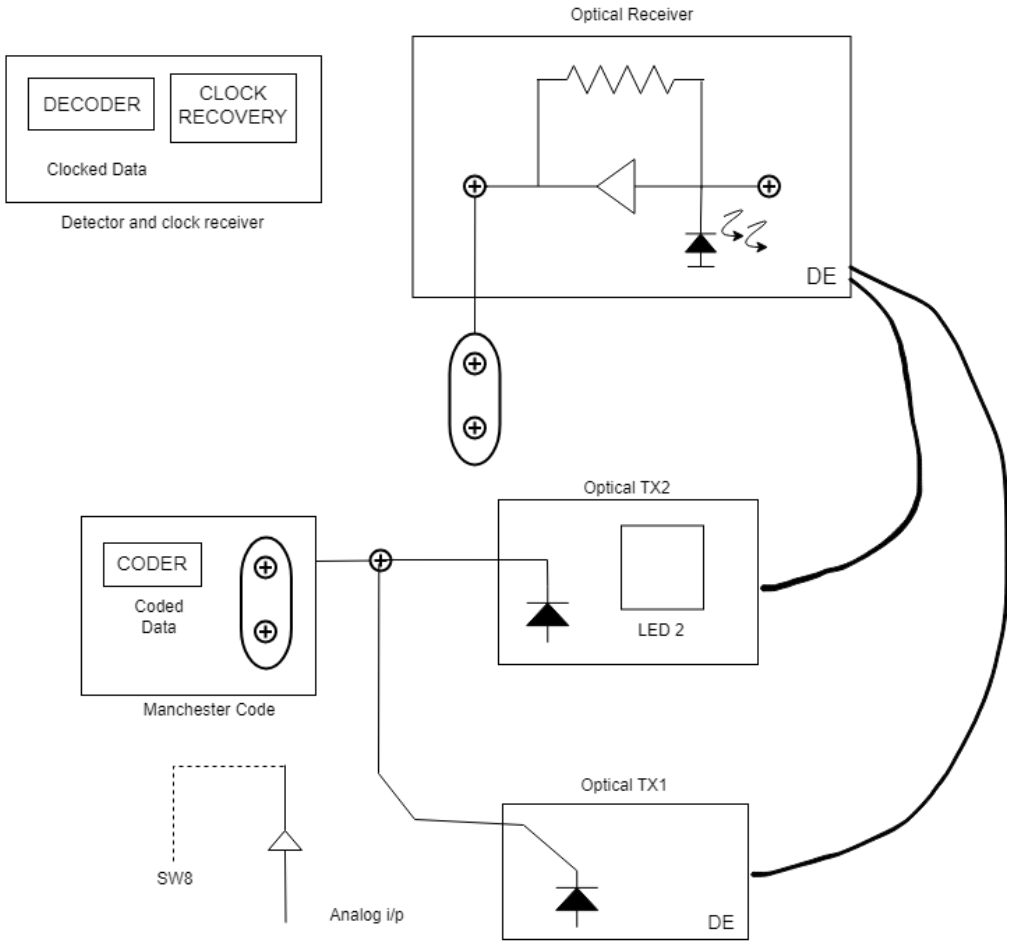
Bending loss depends on the twist that occurs in the optical fiber over the distance which it covers.

Connector loss occurs at the junction where any twofibers get poorly coupled.Any misalignment in the coupling junction increases the connector loss.

BLOCK DIAGRAM:



KIT DIAGRAM:



PROCEDURE:

1. Make the connectionsas shown.
2. Set the switch SWB in analog.
3. Ensure that the shortly plugs from coded data shorting links to it in the Manchester coding.
4. Derive $1V_{P-P}$, 10 KHz, sine wave at P_{11}.
5. Setup the 1m optic fiber.
6. Observe reading in oscilloscope.
7. Repeat the same thing using 3m cable.

OBSERVATION:

Input Voltage V_{in} (V)	Output Voltage V_{out} (V)		Attenuation (Np/m)	Attenuation (dB/m)
	L = 1 m	L= 2 m		
1				
2				
3				
4				
5				

MODEL CALCULATION:

RESULTS AND DISCUSSION:

Expt. No. :

Date :

NUMERICAL APERTURE DETERMINATION OF OPTICAL FIBER

OBJECTIVE:

To measure the numerical aperture (NA) of an optical fiber cable.

SOFTWARE USED:

- OFT Kit
- Numerical Aperture Measurement Kit

THEORY:

Inoptics the numerical aperture of an optical system is a dimensionless number that characterises the range of angle over which the system can accept or reject signals. The numerical aperture of an optical system such as objective lens is given by

$$NA = n \sin \theta$$

The numerical aperture is generally measured with respect to a particular point or image point and will vary as the point is moved. A lens with large NA will be able to visualize finer details than a lens smaller NA.

For NA less than 0.5, the divergence between two, the approximation and full loss than 10% this approximation breaks down.

PROCEDURE:

1. Connect the circuit diagram.

2. The distance between optical fiber and screen is kept constant.

3. Repeat the above steps for various values of ‘d’ and measurable diameter.

4. Calculate the radius of the protected light.

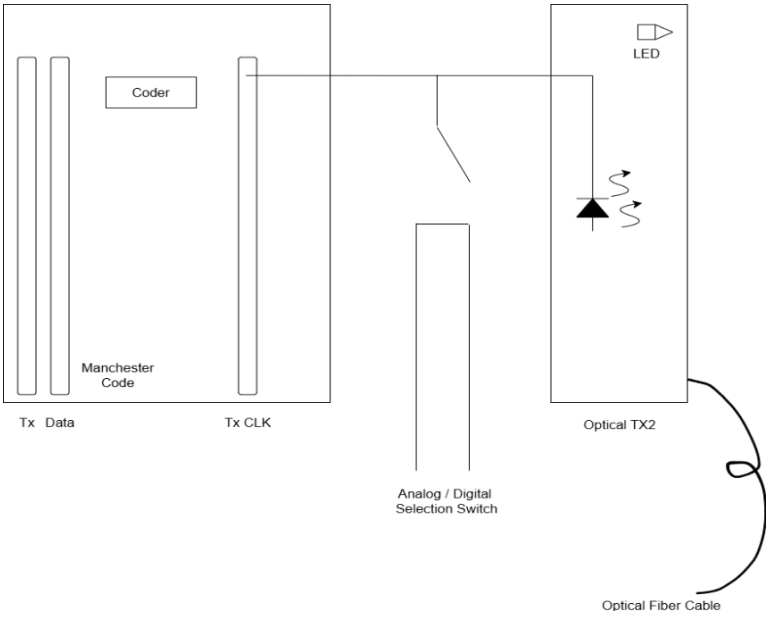
5. Numerical aperture is calculated by:

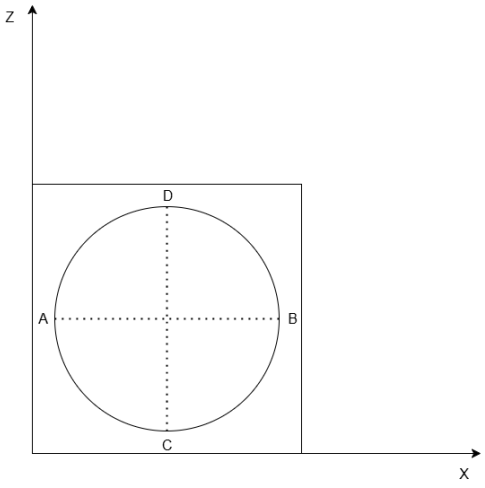
$$NA = \sin \phi_A = \frac{x}{\sqrt{x^2 + d^2}}$$

where x is the radius of the protected light d is

the distance between screen and fiber tip

CONNECTION DIAGRAM:





OBSERVATION:

Distance 'd' between Screen and Fiber end (d) (cm)	The axes of Light on Screen		Radius of Screen $x = \frac{AB+CD}{4}$	$NA = \frac{x}{\sqrt{x^2 + d^2}}$
	AB	CD		

CALCULATION:

RESULTS AND DISCUSSION:

Expt. No. :

Date :

STUDY OF CYLINDRICAL MODE STEP INDEX / GRADED INDEX FIBER

OBJECTIVE:

The objective is to observe lower mode order linear polarised (LP) mode.

EQUIPMENT REQUIRED:

- Laser source (633nm – 1mm)
- Source to Fiber Coupler
- Single mode fiber
- Fiber holding stand
- Opaque Screen

THEORY:

The central spot carries 95% of the intensity for the laser beam with a Gaussian profile. $I = I_0 e^{-\left(2\left(\frac{r}{\omega}\right)^2\right)}$ in the base of a natural logarithm. The definition of a radius of a Gaussian beam intensity has dropped to $\frac{1}{e^2} = 0.135$ times its peak value. If this radius is called spot size, the spot diameter is 'ω'.

Spot diameter (d) micron = final length of the laser (f) Laser Beam full divergence angle (DA)

$$N_{Arrays} = \frac{Laser\ Beam\ Diameter\ (B.D.)}{2 \times lens\ focal\ length\ (f)}$$

If $N_{Arrays} \leq NA$ fiber and spot diameter (d) then all of the laser light will be coupled into the fiber.

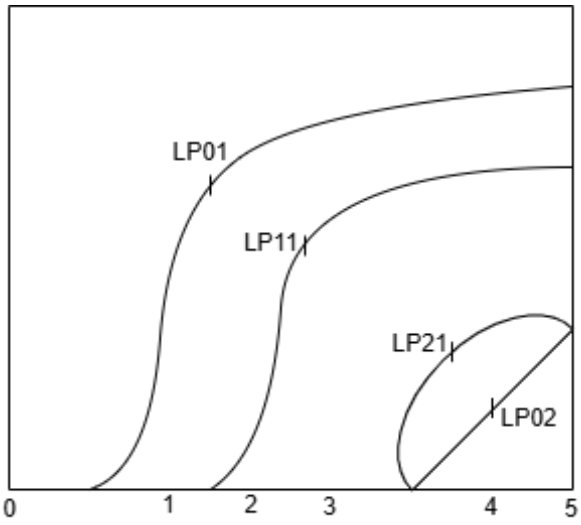
PROCEDURE:

1. Keep the optical breadboard onto original and flat table surface so that it will not toggle.
2. Fix the prefitted cylindrical head of the He-Ne laser source on the surface of the bread board from the bottom side with the help os screws provided with it. Confirm the rigidity of the mount.
3. Fix the laser to the fiber coupler mount onto the bread board with base plate orientation of it towards the He-Ne laser exit.

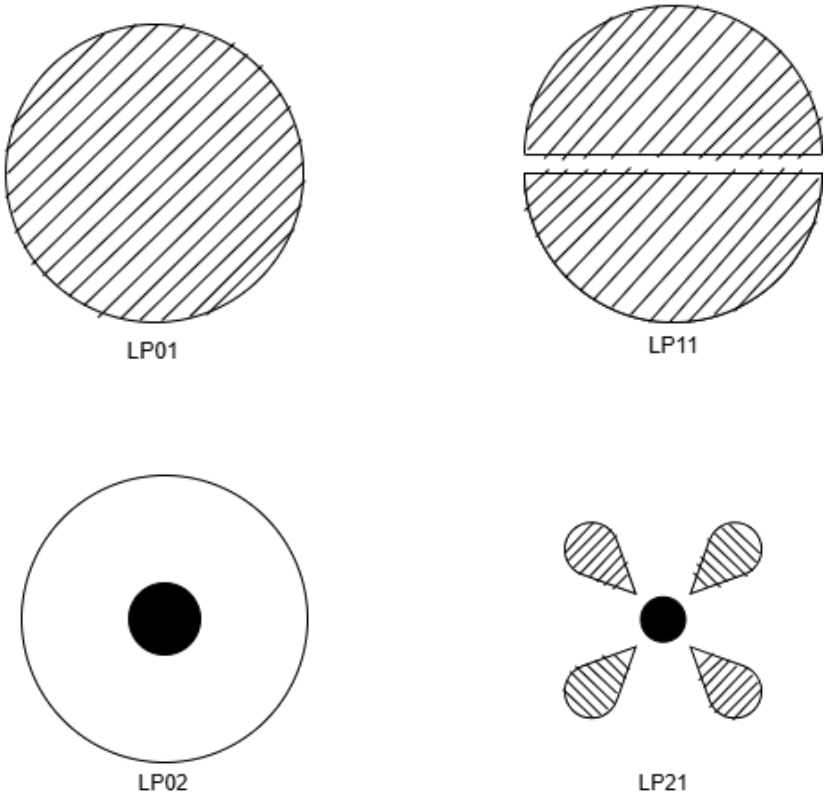
4. Turn on the He-Ne laser and locate the beam spot onto the central portion of the laser fiber coupled lens assembly by adjusting the vertical and horizontal slots.
5. Now look for the back reflection of the He-Ne laser spot from the rod lens of the coupler. In case it is found that the backspotis away from the exit of the cylindrical laser beam head of the laser, adjust the back reflected spot going back in exit hole by slowly moving the four screws provided for the laser mount.
6. Confirm the central alignment of the laser beam at the exit of the laser beam at the laser fiber coupler by putting a white card sheet and zooming the spot onto it.
7. Put the multimode optical patch chord onto the laser fiber coupler exit and fix the other end of the fiber in the fiber holding stand by moving the grub screws produced with the holder.
8. A bright laser beam coming out of the fiber is observed.
9. A round shape spot with laser pattern appears on screen.
10. Once you observe the multimode pattern, change the multimode patch chord with single mode chord.
11. As soon as you place the single mode patch and you will see the true pattern of various pattern out of the screen.

DIAGRAM:

Low Order Linearly Polarised Modes of Optical Fibers:



Irradiance pattern of some Low Order Linearly Polarized Modes:



RESULT:

Depending on the V-number and wavelength fiber can be characterized as single mode or multimode. Some lower order LP modes are observed with fiber whole V-number is 4.91. The total number of modes is 12.

Expt. No. :

Date :

STUDY OF FIBER OPTIC ANALOG AND DIGITAL LINK

AIM:

To study the fiber optic analog and digital link with the help of 1310 nm laser diode.

SOFTWARE USED:

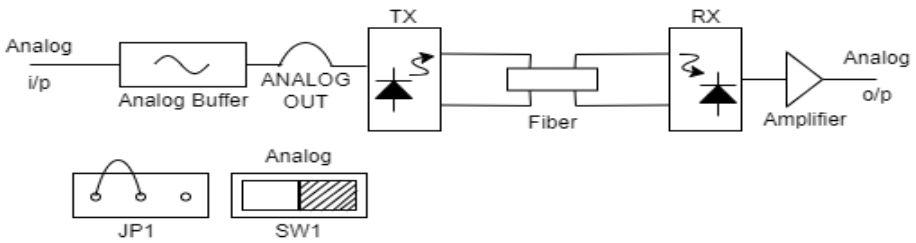
- Fiber Link E-Kit
- Glass Fiber Cable with SI connector
- Patch chords
- CRO
- Signal Generator Power Supply

THEORY:

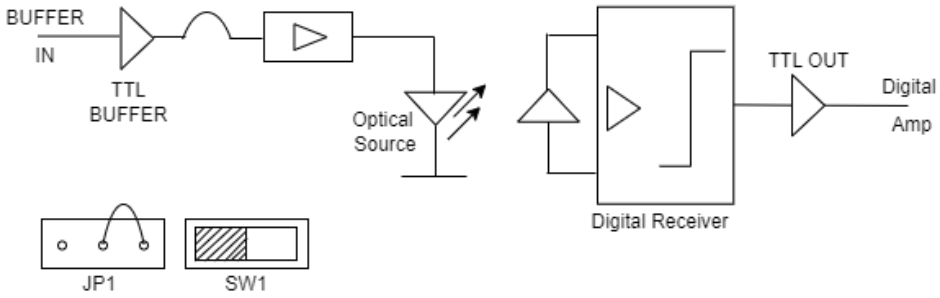
Fiber optic link can be used for transmission of digital as well as analog signal. Basically, a fiber optic link contains three main elements, a transmitter, a receiver and an optical fiber. The transmitter takes the input signal in electrical form when transforms it into optical energy.The optical fiber is the medium which carries this energy to the receiver.At the receiver, light is converted back into electrical form with same pattern as fed to transmitter.Analog systems are mainly used to transmit video signals, for applications of telecontrol systems, videoconferences etc. Digital systems are suitable for the transmission of TTL digital signals.The TTL signals modulate the intensity of light emitted by laser diode.

KIT CONNECTION DIAGRAM:

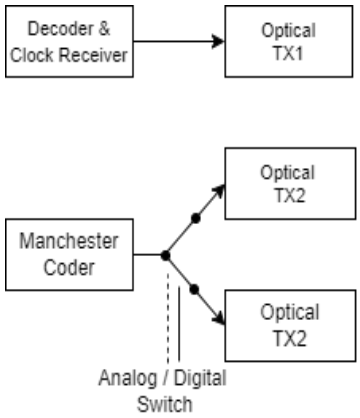
Setting Up Fiber Optic Analog Link:



Setting of Digital Link Fiber Optic Communication:



BLOCK DIAGRAM:



PROCEDURE:

1. Confirm that the power switch is in OFF position and then connected to the kit.
2. Make the jumper settings and connection as shown in diagram.
3. Keep the pot P_5 in anticlockwise rotation. It is used to control the intensity of laser diode.
4. Connect the external signal generator to analogIN port of analog bufferedBUFFER IN
5. port of TTL and keep the signal generator in analog mode and generate a sine wave of f = 1 MHz, 1 V_{P-P} .
6. Switch ON the power supply and signal generator.
7. Check the output signal of the analog buffer port of the ANALOG OUT or TTL buffer port of the BUFFER OUT. It should be same as that of the output. Connect the fiber between laser diode and detector.
8. Observe the output signal from detector at DETECTOR OUT port on CRO by adjusting P_5 till you get the input or from the detector of TTL OUT port in the receiver section on CRO with the reproduction of original signal.
9. Vary the frequency of the input signal and observe the output response. Observe the variation of duty cycle of the signal.

OBSERVATION:

Analog Link : Voltage vs Frequency Response

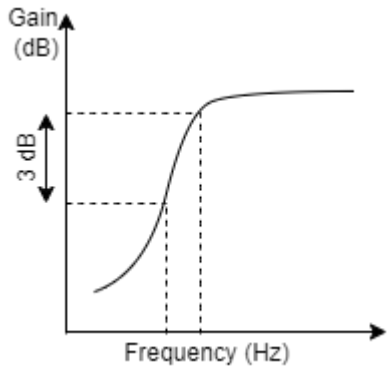
V_{in} = 1 V

Input Frequency (KHz)	Output Voltage V_{out} (V)	Gain (dB)
1		
50		
90		
150		
200		
300		
600		

Digital Link:

	V_{pp} (V)	T_{ON} (μs)	T_{OFF} (μs)
CLK			
DS without noise			
DS with noise			

MODEL GRAPH:



RESULTS AND DISCUSSION: